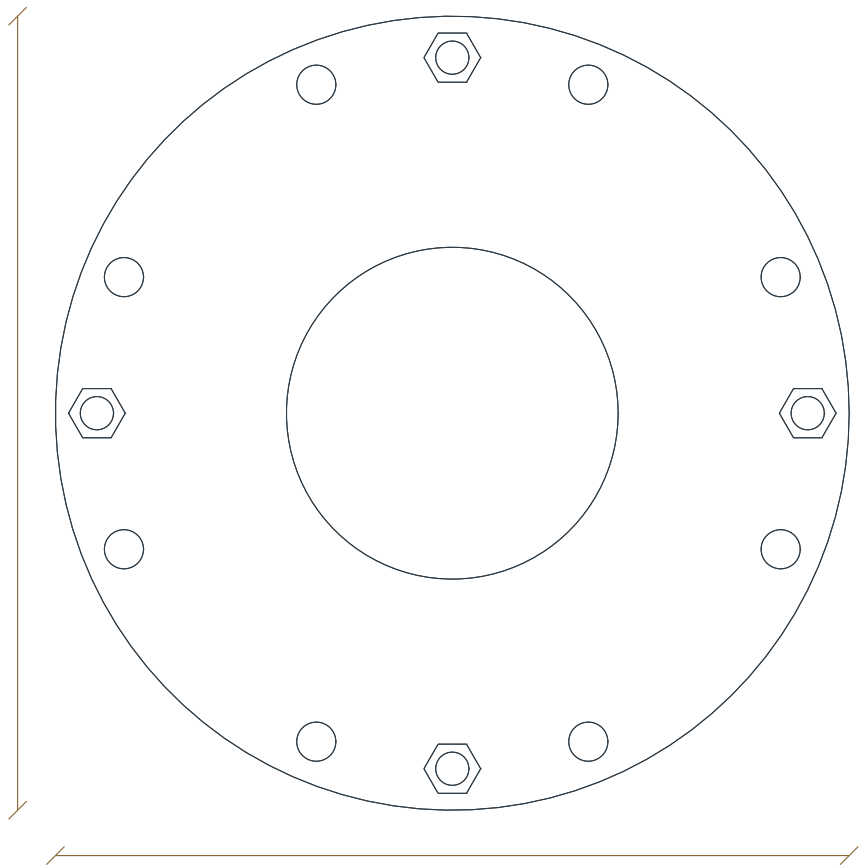


A16-R101-base-adapter

Frame world | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

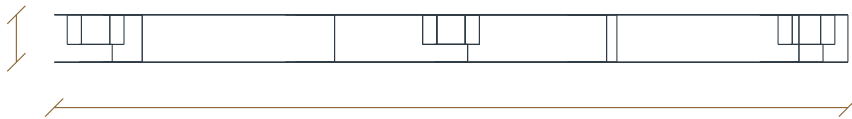
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



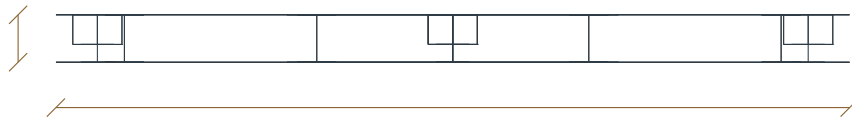
W 134.000 / H 134.000 / scale 0.7836  
CAD lower-left XY: -67.000, -67.000

XZ / positive axes right and up



W 134.000 / H 8.000 / scale 0.7836  
CAD lower-left XZ: -67.000, 58.000

YZ / positive axes right and up



W 134.000 / H 8.000 / scale 0.7836  
CAD lower-left YZ: -67.000, 58.000

OD134; PCD120 8xD6.6 phase22.5; 4xD5.6 at cardinal phase with top-entry AF8.3 depth5 captive M5 nuts; boreD56. Install above B06 neck.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

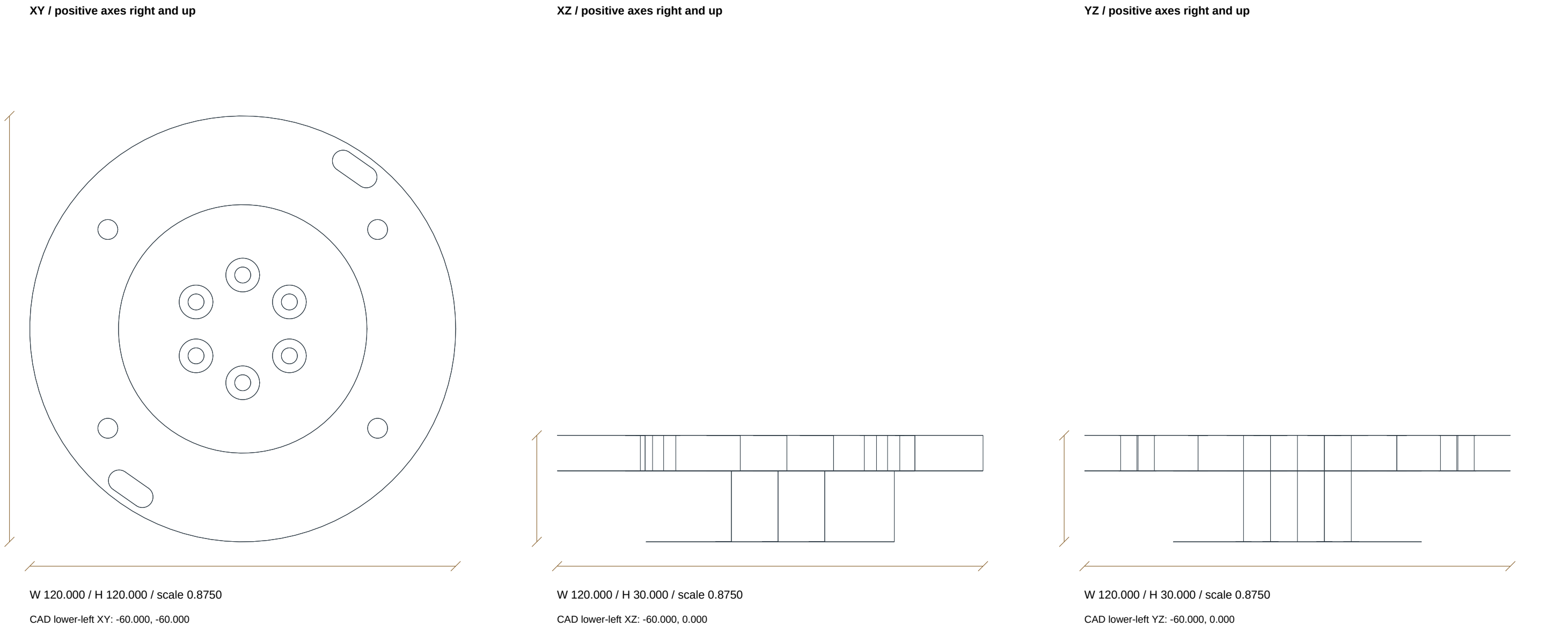
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-R103-J1-output-pedestal

Frame J1.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



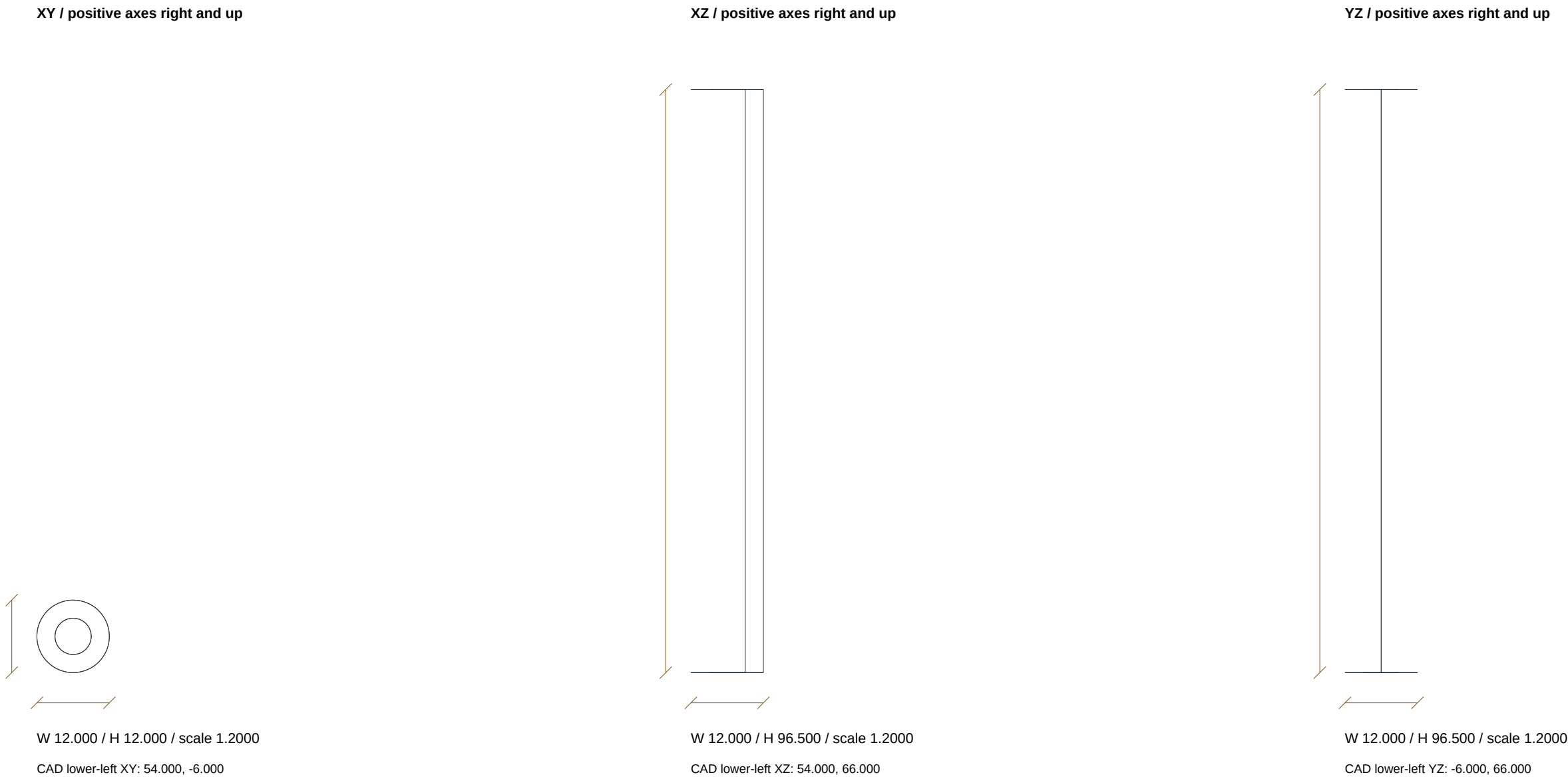
One-piece lathe/mill geometry; broad upper plate DOWN, motor contact UP for supported unloaded fit. D70x20 stem, D120x10 upper plate; six recessed M4x25.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-R104-spacer-1

Frame world | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



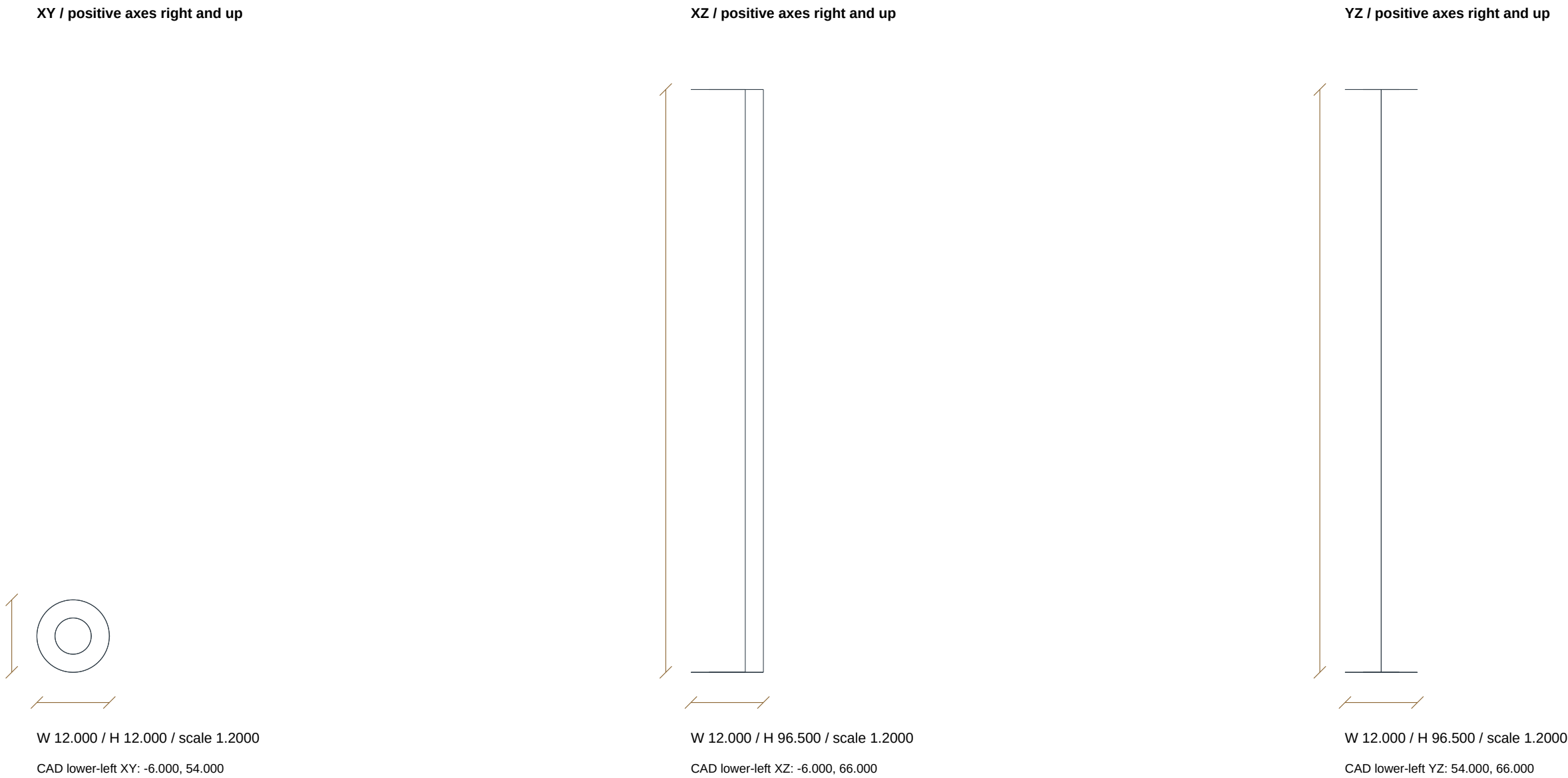
Purchase metal spacer for prototype as well; do not print as a loaded column. Tube covers captive nut and bears on plate around the pocket.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-R104-spacer-2

Frame world | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Purchase metal spacer for prototype as well; do not print as a loaded column. Tube covers captive nut and bears on plate around the pocket.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

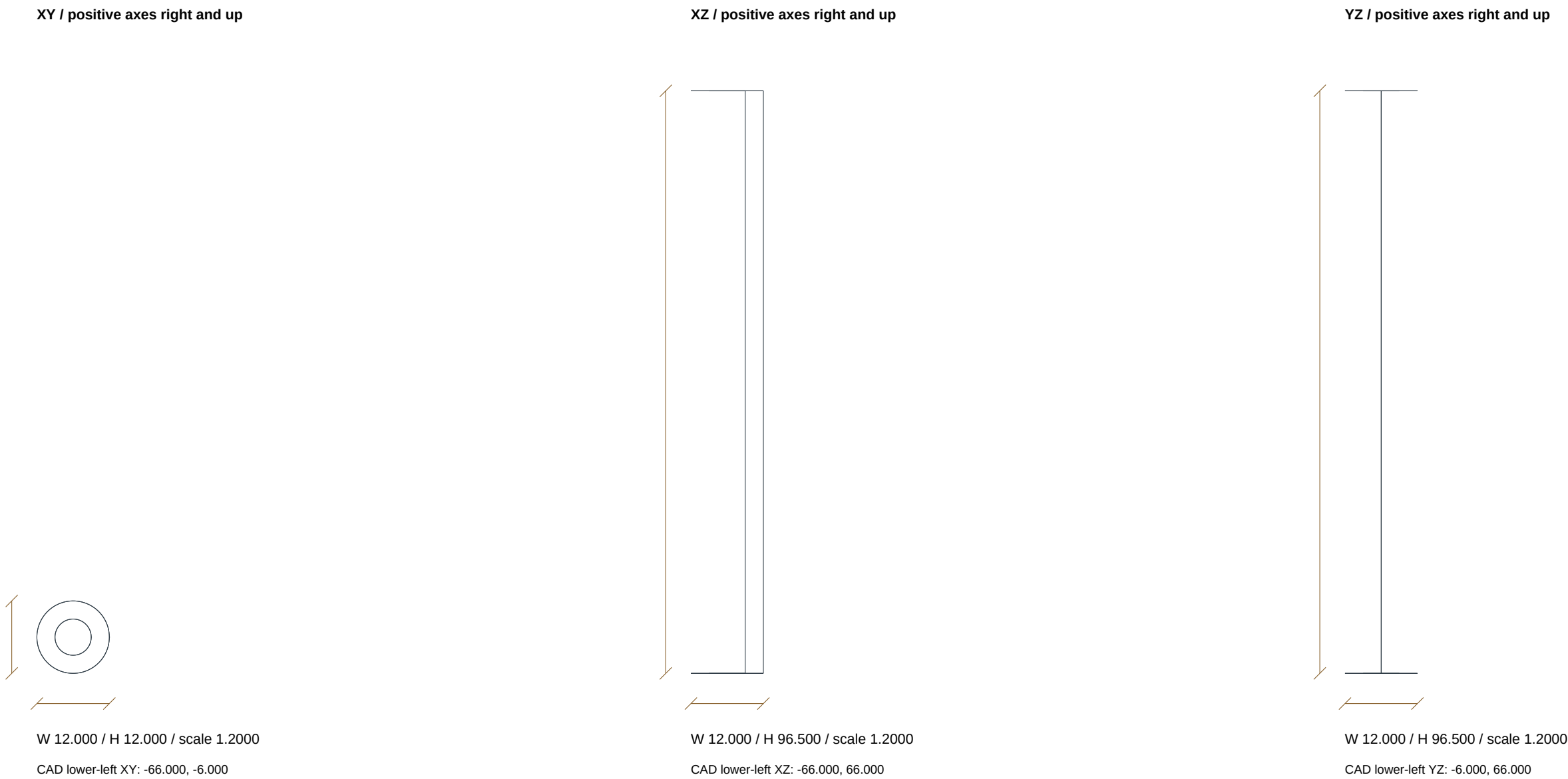
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

# A16-R104-spacer-3

Frame world | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



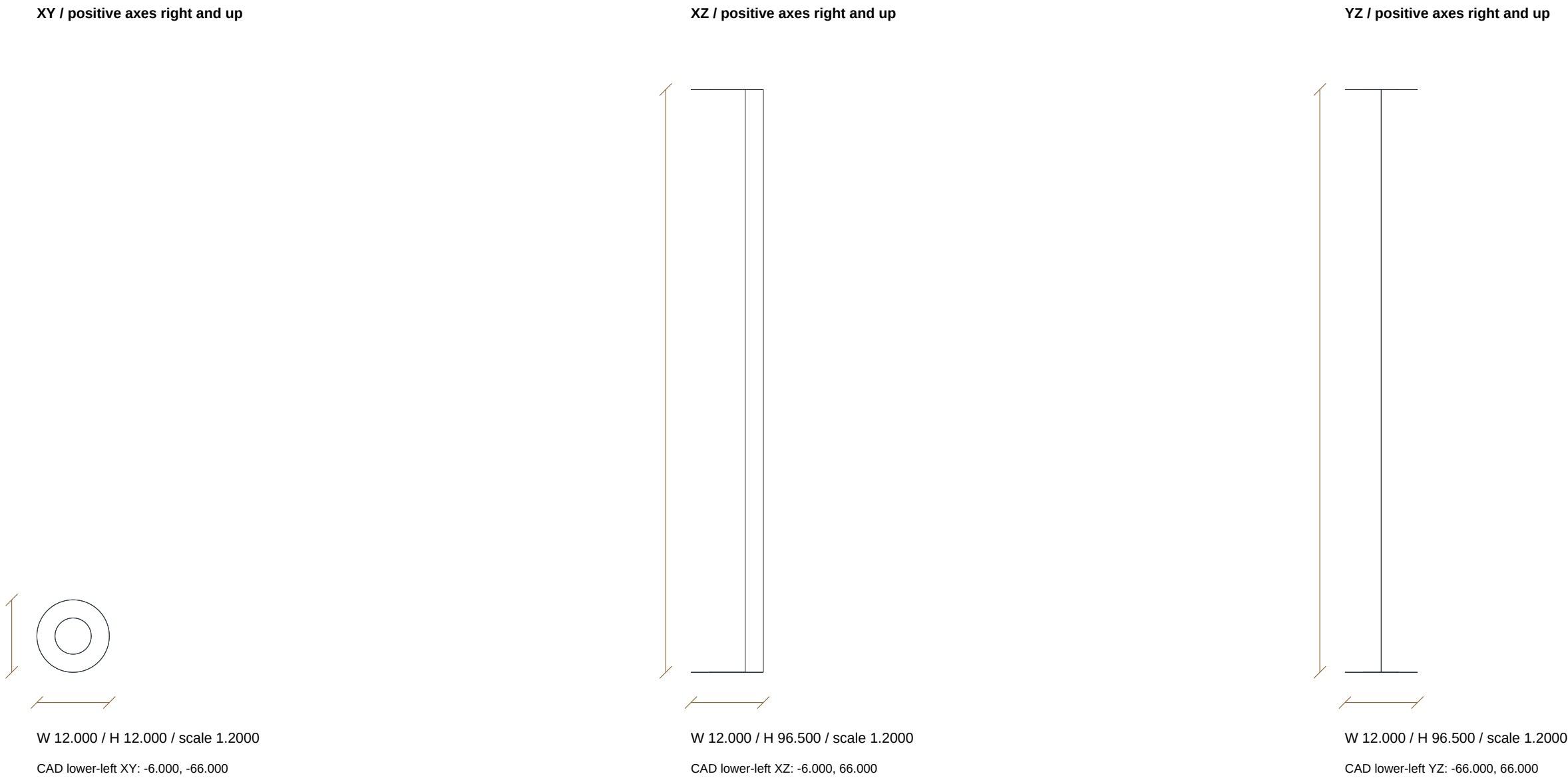
Purchase metal spacer for prototype as well; do not print as a loaded column. Tube covers captive nut and bears on plate around the pocket.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-R104-spacer-4

Frame world | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Purchase metal spacer for prototype as well; do not print as a loaded column. Tube covers captive nut and bears on plate around the pocket.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

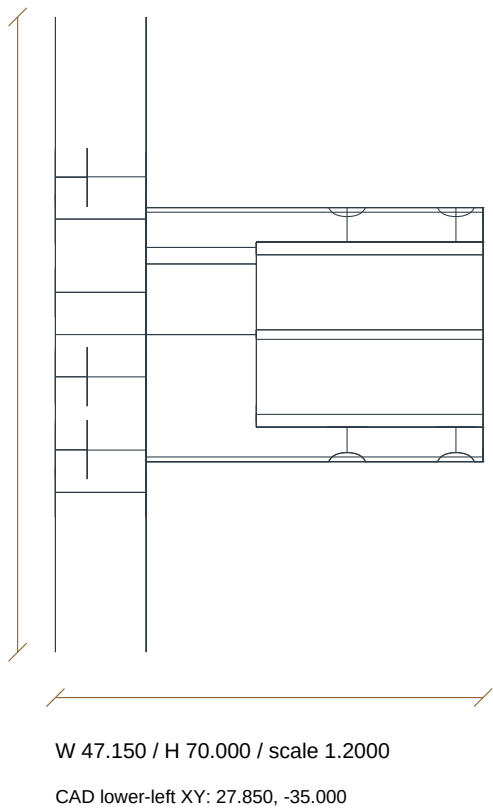
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-S103-upper-proximal-socket

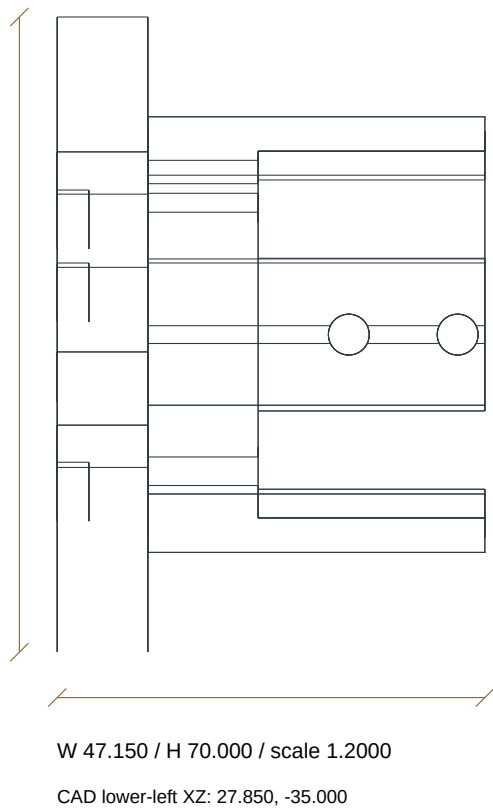
Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

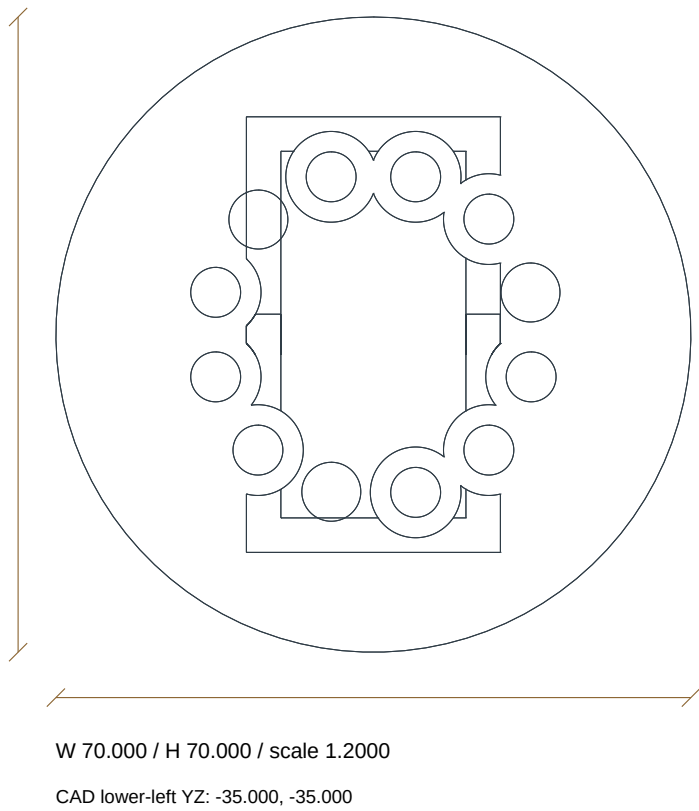
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



220mm stock20x40x2 tube beginsX50. Fit gap0.2 per side;2xM4 transverse clamp through tube; output bolts before tube.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-S105-upper-stock-tube

Frame J3.rotor | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Catalogue material assumption only, not strength certification. Tube is purchased metal in supported fit.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

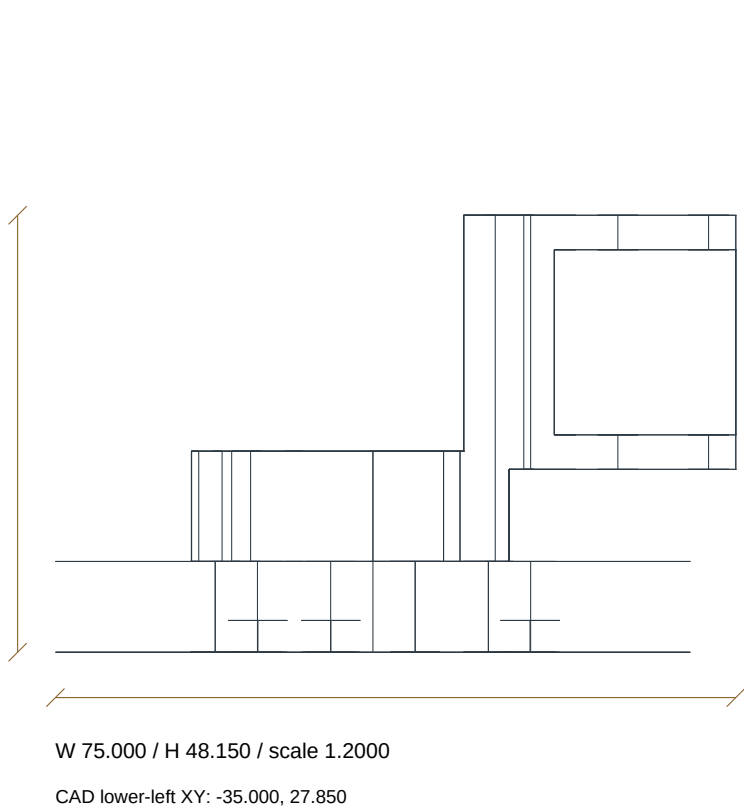
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-S106-fore-proximal-socket

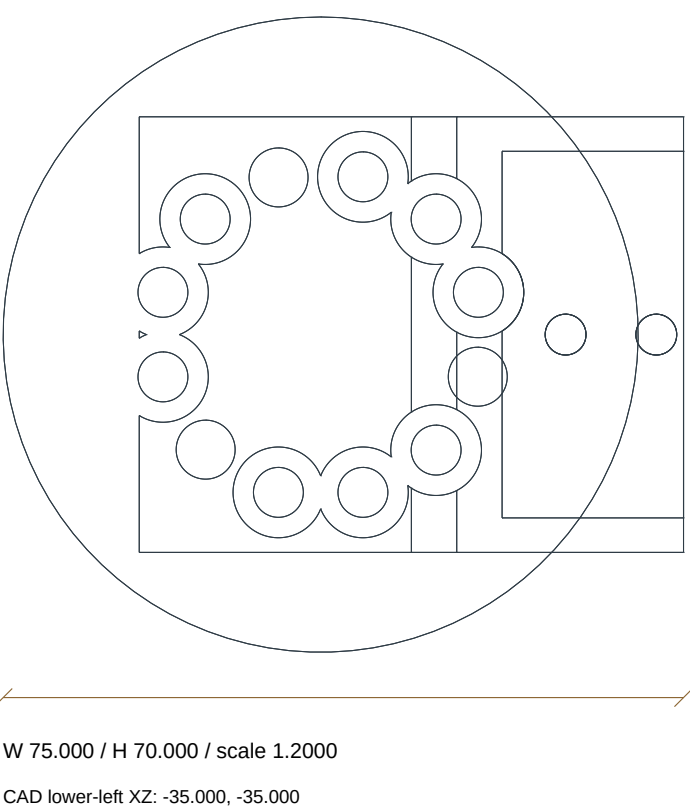
Frame J4.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

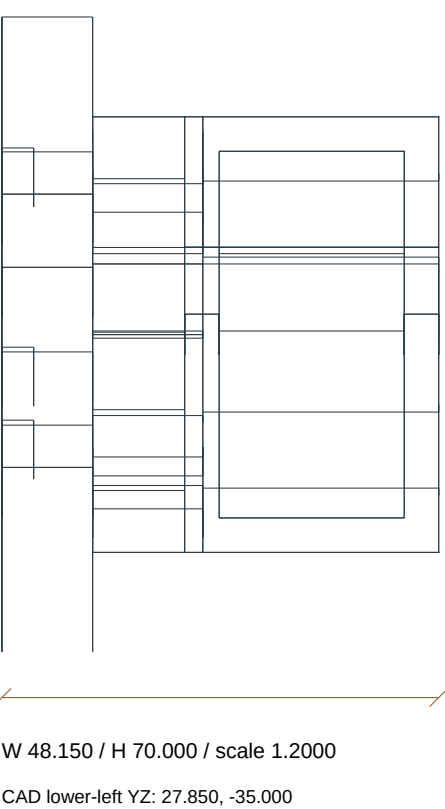
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



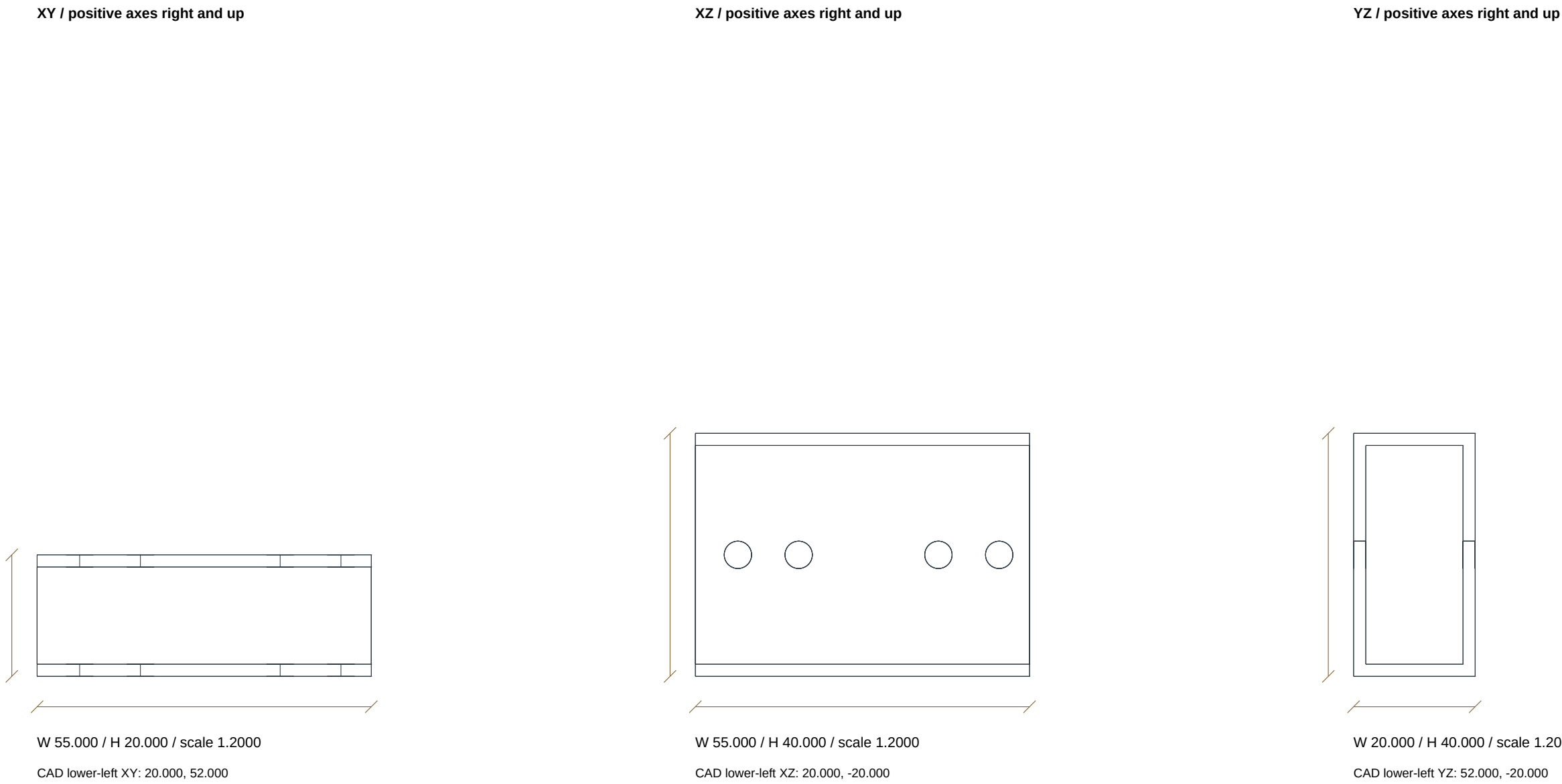
Ordinary elbow L/socket; tube centreY62. Bolt J4 output before installing tube.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-S108-fore-stock-tube

Frame J4.rotor | purchased\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Fore axis spacing185 unchanged; fixed rear spine completes span outside wrist pitch sweep.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-S110-wrist-yaw-roll-L

Frame J6.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



J7 front ring/cartridge planes retained;6x cage through boltsPCD70; annular screw-head sweep relief, no extra mechanism.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

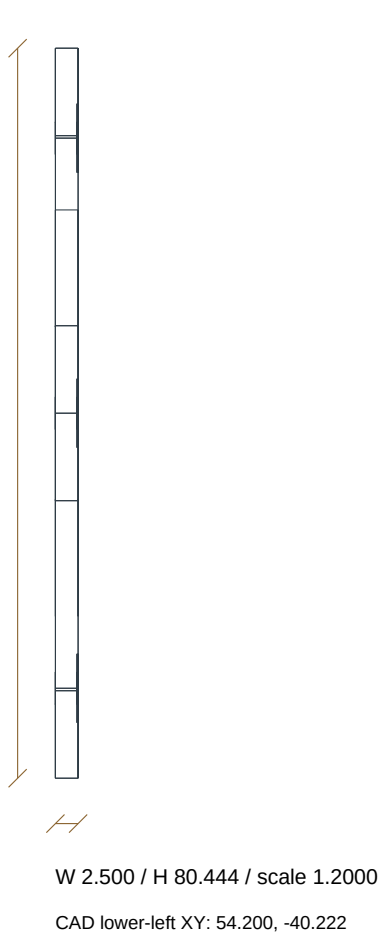
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-J7-102-bearing-retainer

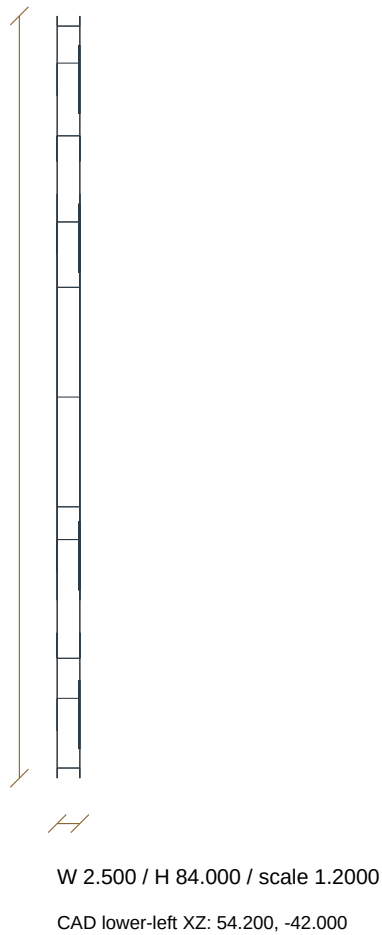
Frame J7.fixed | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

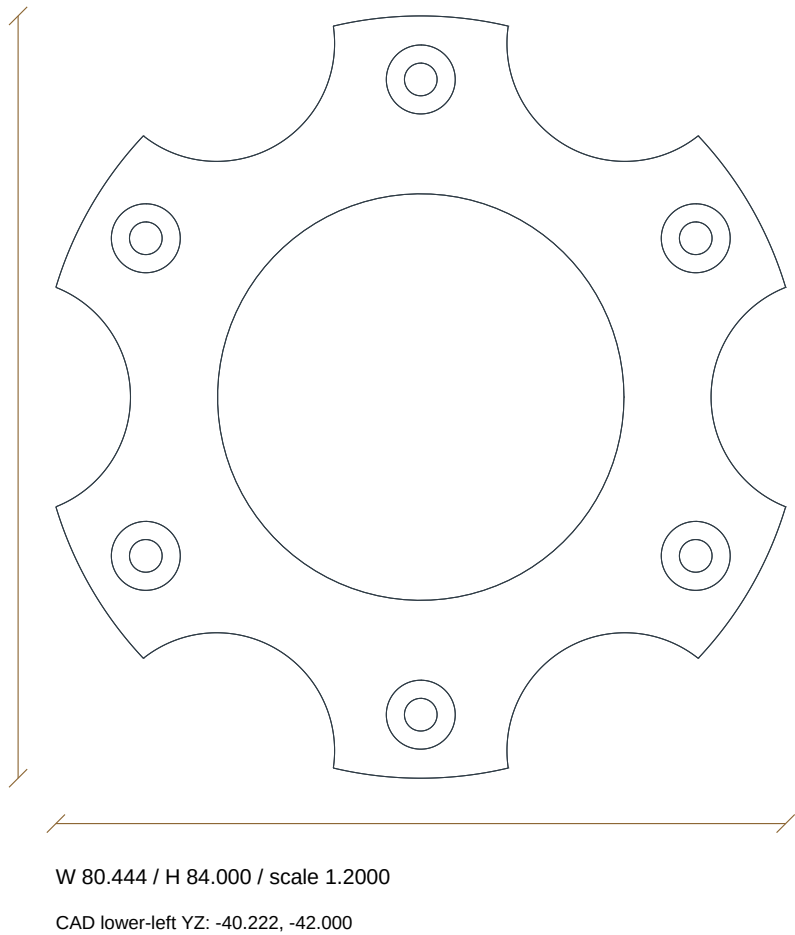
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

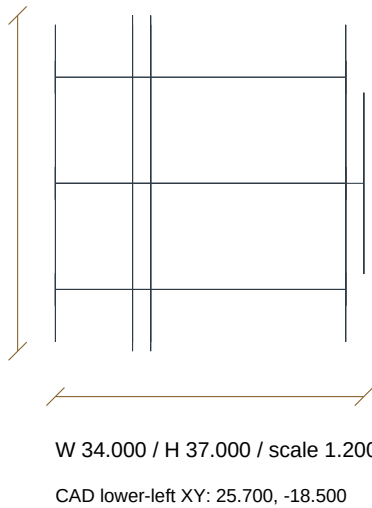
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-J7-103-output-journal

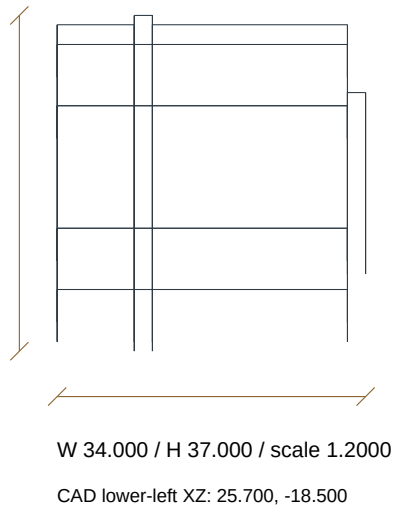
Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

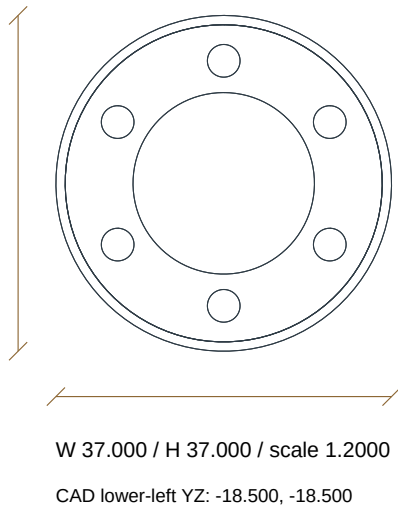
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

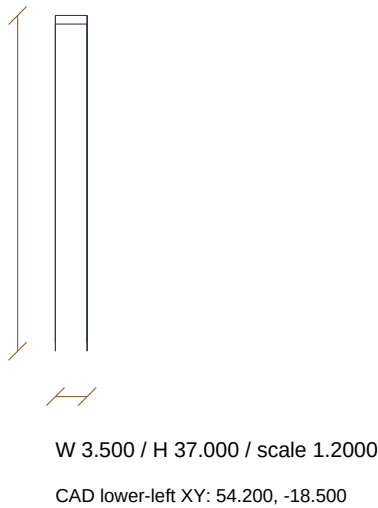
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-J7-104-inner-spacer

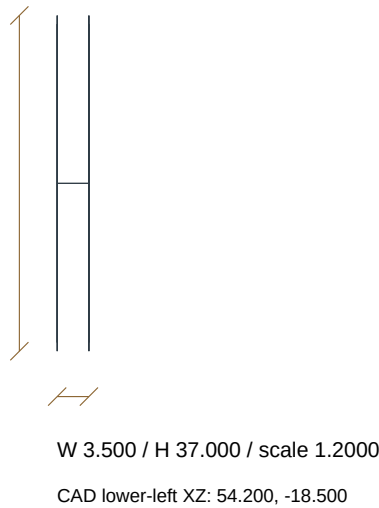
Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

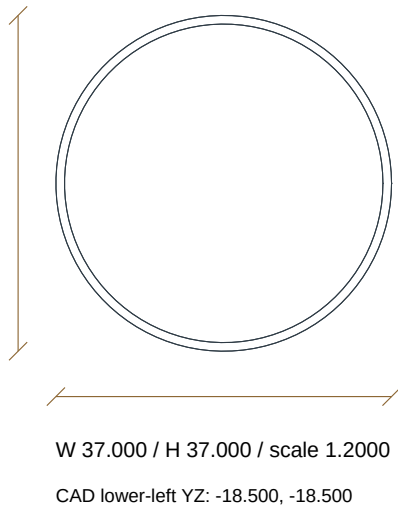
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

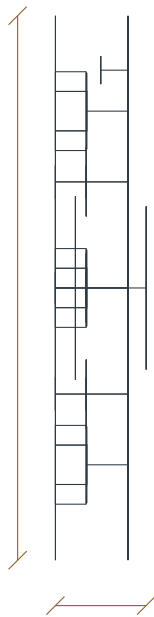
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-IF-107-piloted-flange

Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



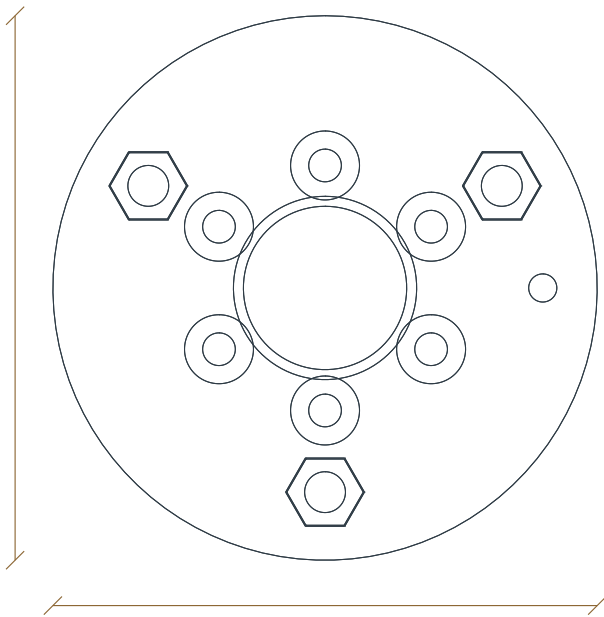
W 10.000 / H 60.000 / scale 1.2000  
CAD lower-left XY: 57.700, -30.000

XZ / positive axes right and up



W 10.000 / H 60.000 / scale 1.2000  
CAD lower-left XZ: 57.700, -30.000

YZ / positive axes right and up



W 60.000 / H 60.000 / scale 1.2000  
CAD lower-left YZ: -30.000, -30.000

Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

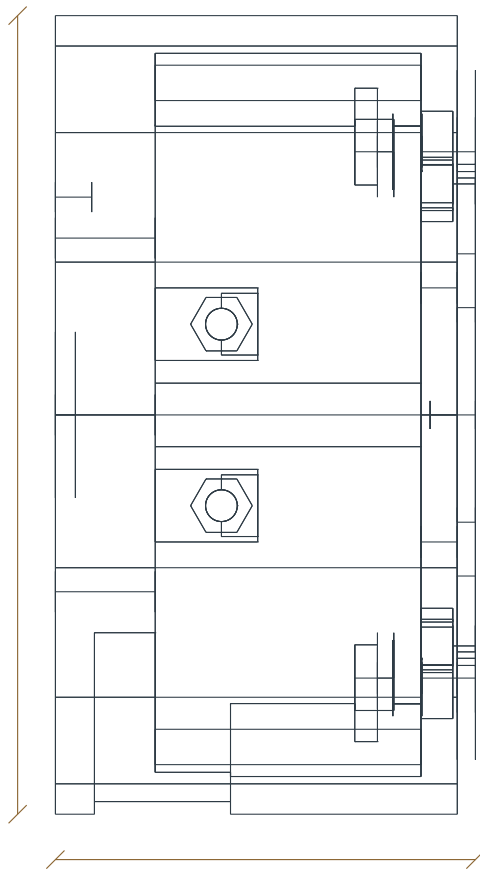
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-IF-301-recessed-carrier

Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

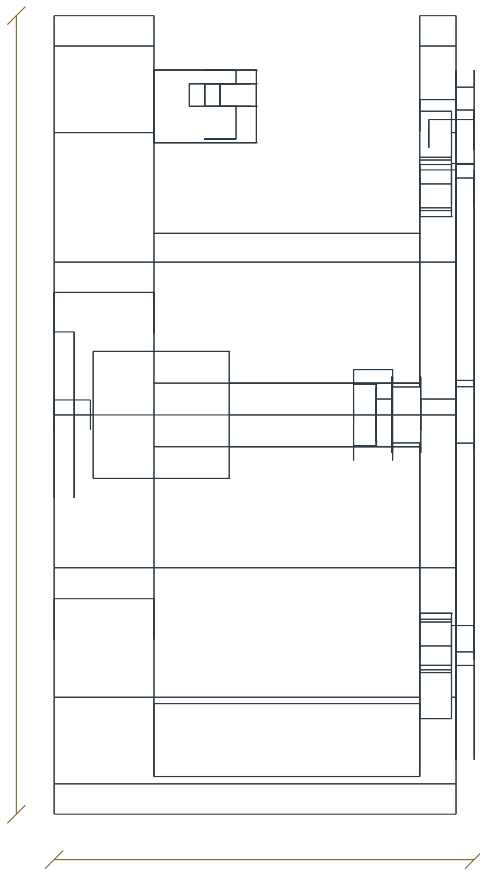
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



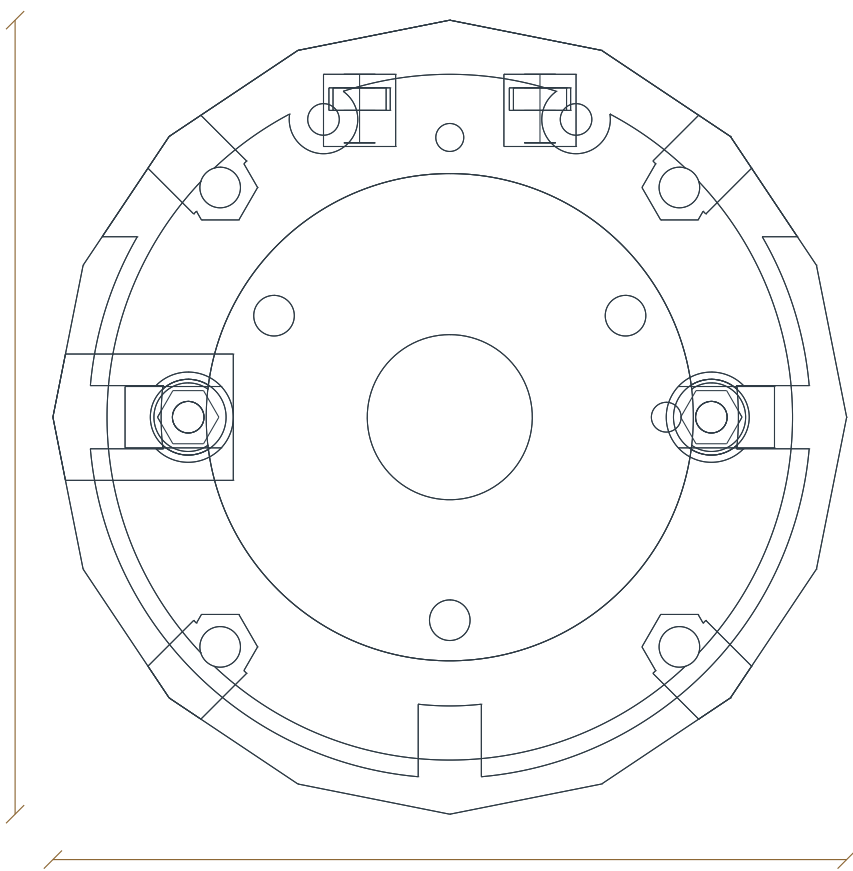
W 46.300 / H 88.000 / scale 1.2000  
CAD lower-left XY: 65.700, -44.000

XZ / positive axes right and up



W 46.300 / H 88.000 / scale 1.2000  
CAD lower-left XZ: 65.700, -44.000

YZ / positive axes right and up



W 88.000 / H 88.000 / scale 1.1932  
CAD lower-left YZ: -44.000, -44.000

Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

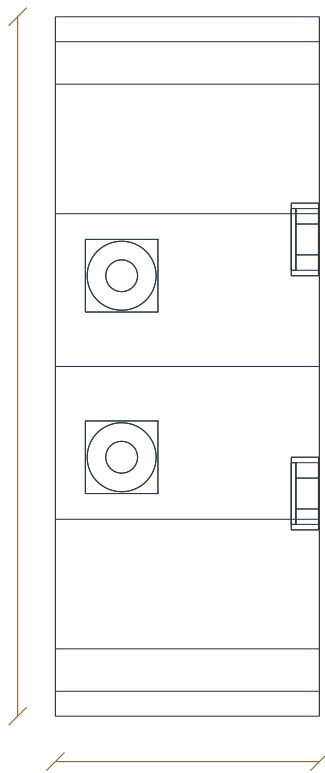
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A13-IF-302-dorsal-cover

Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

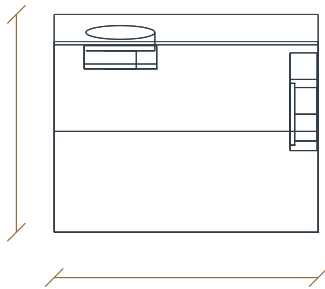
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



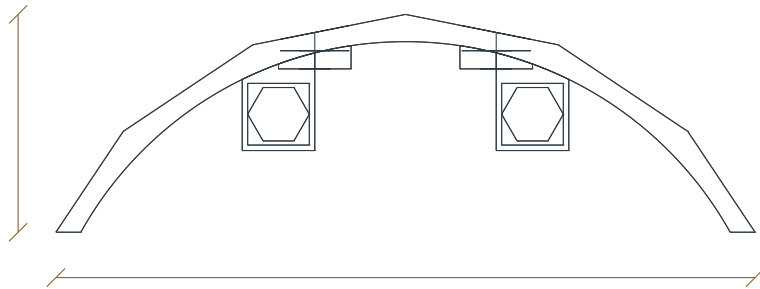
W 29.100 / H 77.076 / scale 1.2000  
CAD lower-left XY: 76.700, -38.538

XZ / positive axes right and up



W 29.100 / H 24.000 / scale 1.2000  
CAD lower-left XZ: 76.700, 20.000

YZ / positive axes right and up



W 77.076 / H 24.000 / scale 1.2000  
CAD lower-left YZ: -38.538, 20.000

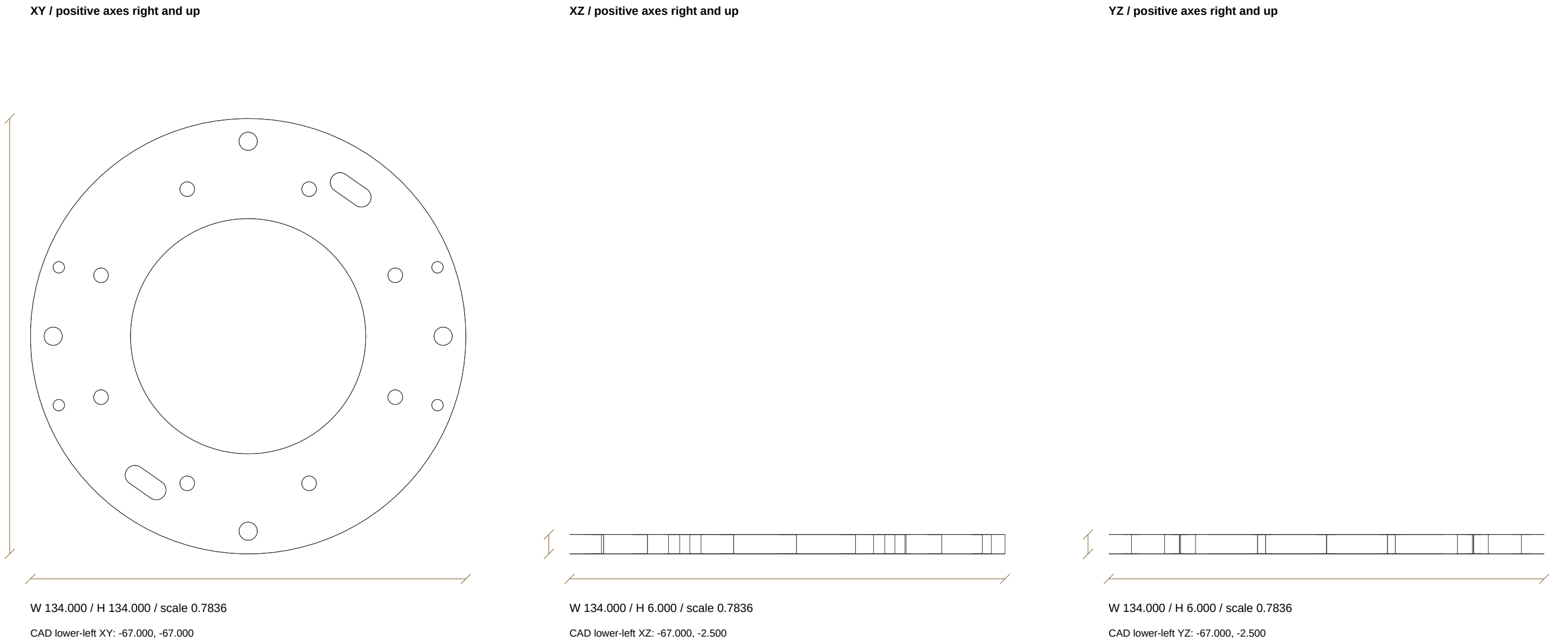
Unchanged valid A13 STEP, no topology cleanup; new arm context, qualification pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C03-root-front-holder

Frame J1.fixed | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



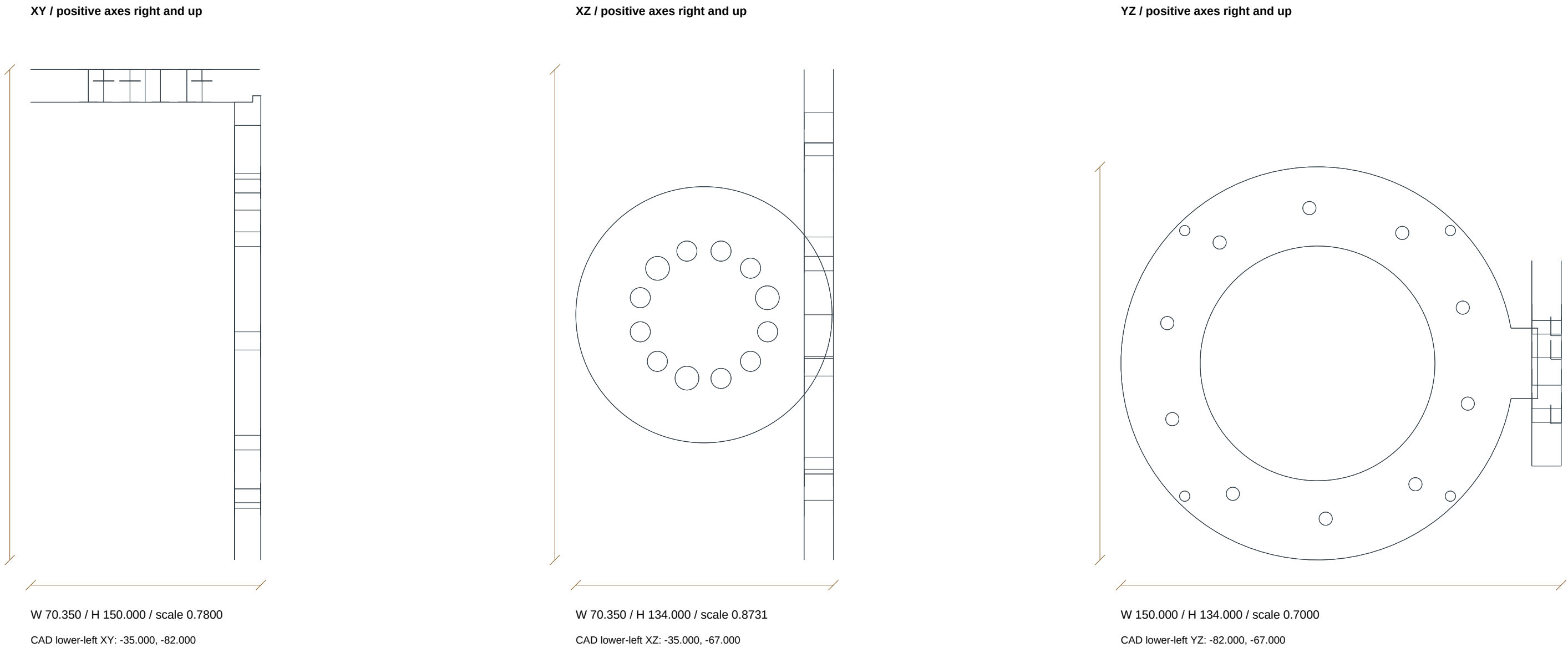
Replacement R102 only: four additionalD3.5 atR62,20/160/200/340deg. X+/-58.2609 lies outside S101 foot+/-55, allowing an outward hex-key access allocation. Native mounts/slots/datums unchanged; old R102 is not simultaneously installed.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C04-J3-mount-plate

Frame J2.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



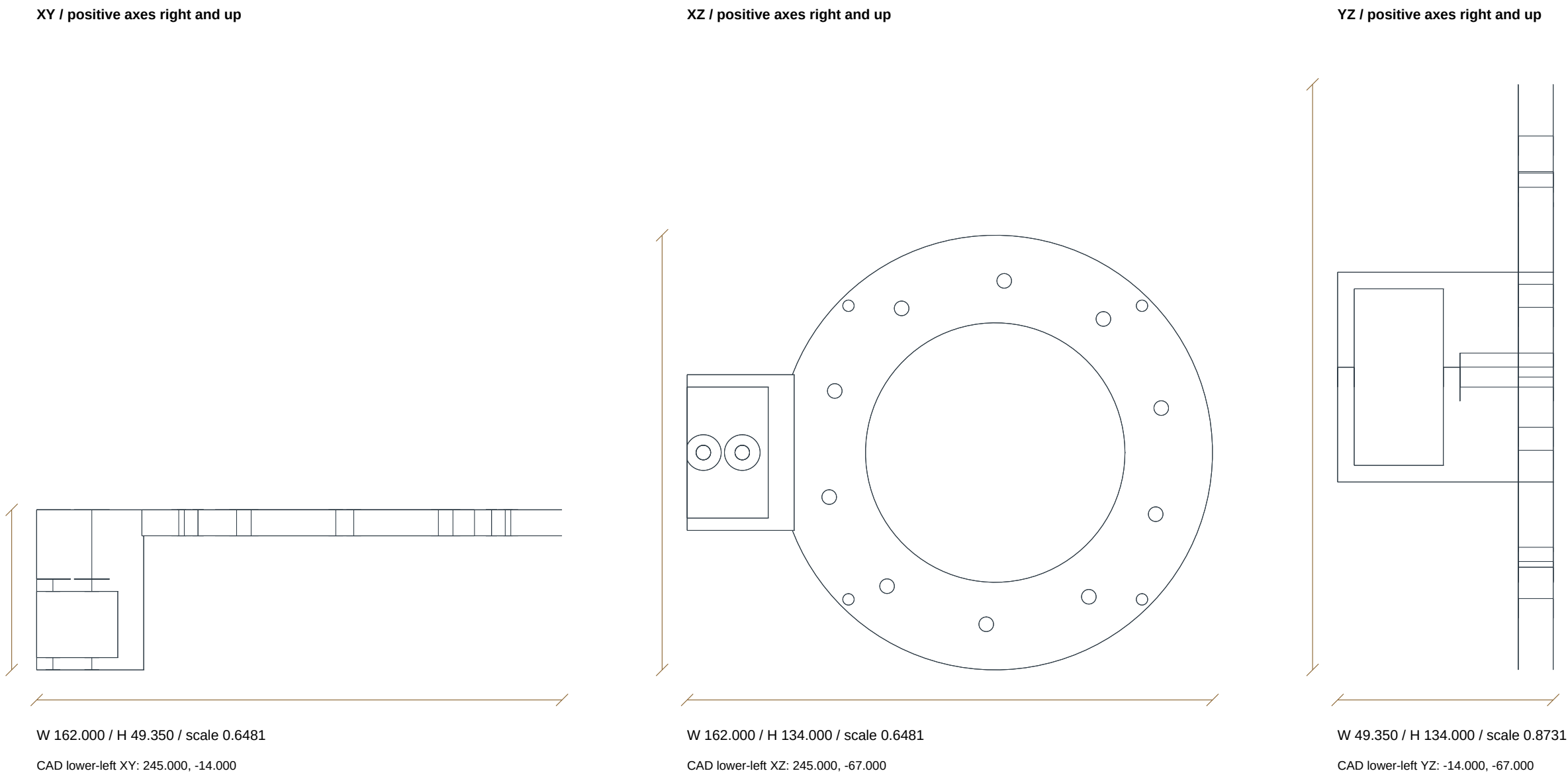
Existing core plate replaced, fourD3.5 atR64; native mounting faces and source hole patterns unchanged. No plastic threads.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C04-J4-mount-plate

Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Existing core plate replaced, fourD3.5 atR64; native mounting faces and source hole patterns unchanged. No plastic threads.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

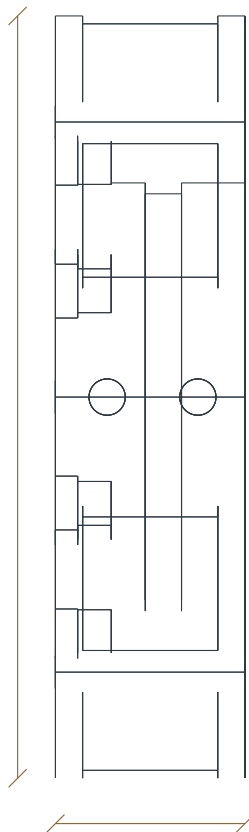
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C05-J7-bearing-housing

Frame J7.fixed | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

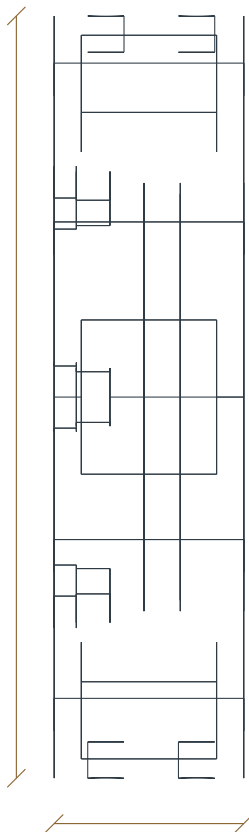
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



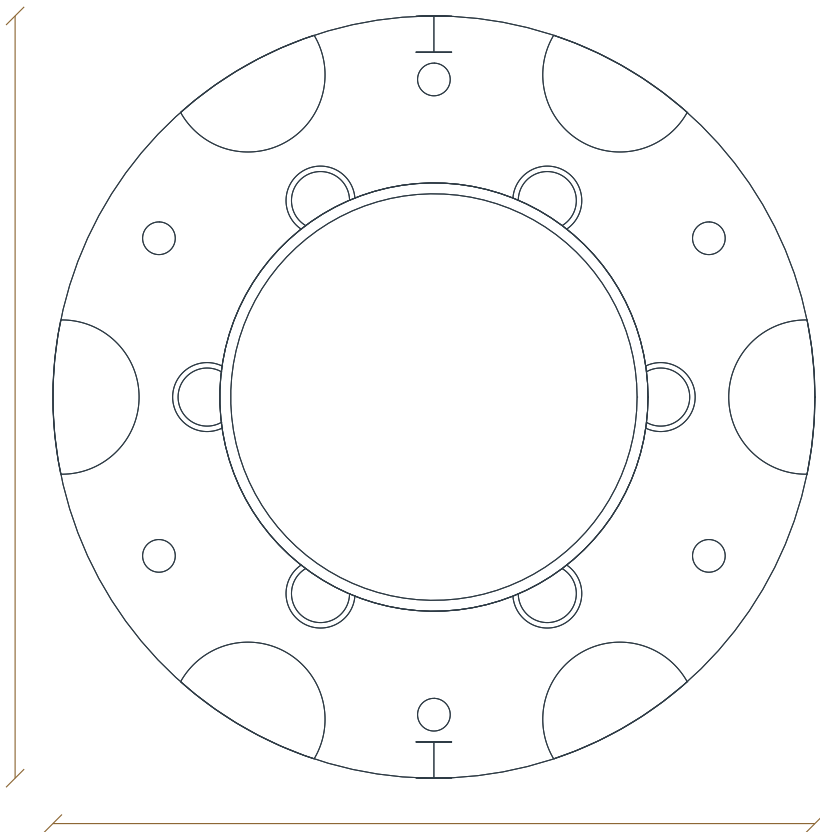
W 20.900 / H 84.000 / scale 1.2000  
CAD lower-left XY: 33.300, -42.000

XZ / positive axes right and up



W 20.900 / H 84.000 / scale 1.2000  
CAD lower-left XZ: 33.300, -42.000

YZ / positive axes right and up



W 84.000 / H 84.000 / scale 1.2000  
CAD lower-left YZ: -42.000, -42.000

Original bearing seats/cage bolts unchanged; fourD4.0x4.0 radial heat-set pilots atX39/49,Y0,Z+/-42. RX-M3Sx4.0; physical bearing/pullout checks pending.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

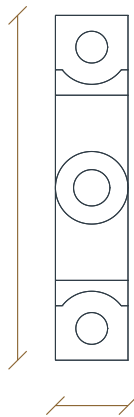
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C06-upper-saddle-115-dorsal

Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

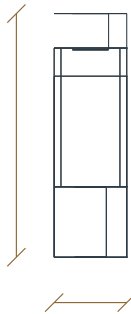
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



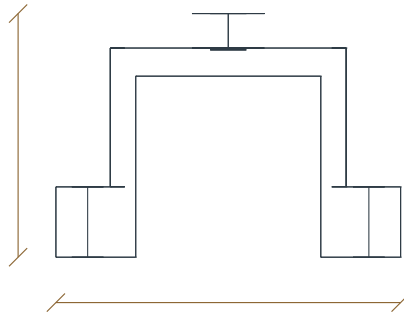
W 8.000 / H 38.000 / scale 1.2000  
CAD lower-left XY: 111.000, -19.000

XZ / positive axes right and up



W 8.000 / H 26.848 / scale 1.2000  
CAD lower-left XZ: 111.000, 0.250

YZ / positive axes right and up



W 38.000 / H 26.848 / scale 1.2000  
CAD lower-left YZ: -19.000, 0.250

Conventional split8mm-long saddle around stock20x40 tube; cavity20.4x40.4, seam0.5. Two ordinary clamp bolts; no stock-tube drilling or cable passage blockage. Tightening/friction requires physical fit.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

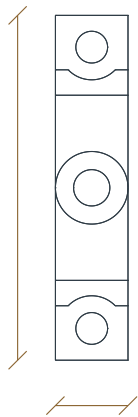
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C06-upper-saddle-115-ventral

Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

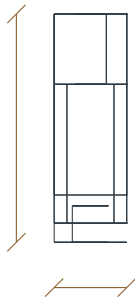
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



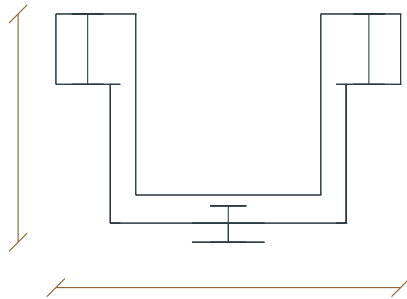
W 8.000 / H 38.000 / scale 1.2000  
CAD lower-left XY: 111.000, -19.000

XZ / positive axes right and up



W 8.000 / H 25.152 / scale 1.2000  
CAD lower-left XZ: 111.000, -25.402

YZ / positive axes right and up



W 38.000 / H 25.152 / scale 1.2000  
CAD lower-left YZ: -19.000, -25.402

Conventional split8mm-long saddle around stock20x40 tube; cavity20.4x40.4, seam0.5. Two ordinary clamp bolts; no stock-tube drilling or cable passage blockage. Tightening/friction requires physical fit.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

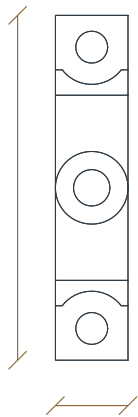
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C06-upper-saddle-200-dorsal

Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

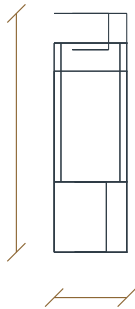
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



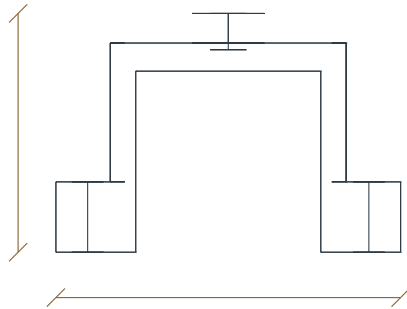
W 8.000 / H 38.000 / scale 1.2000  
CAD lower-left XY: 196.000, -19.000

XZ / positive axes right and up



W 8.000 / H 26.311 / scale 1.2000  
CAD lower-left XZ: 196.000, 0.250

YZ / positive axes right and up



W 38.000 / H 26.311 / scale 1.2000  
CAD lower-left YZ: -19.000, 0.250

Conventional split8mm-long saddle around stock20x40 tube; cavity20.4x40.4, seam0.5. Two ordinary clamp bolts; no stock-tube drilling or cable passage blockage. Tightening/friction requires physical fit.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

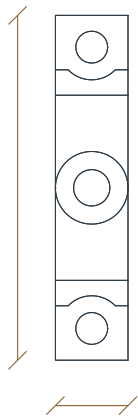
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A16-C06-upper-saddle-200-ventral

Frame J3.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

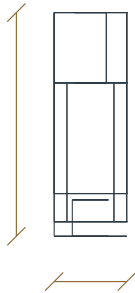
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



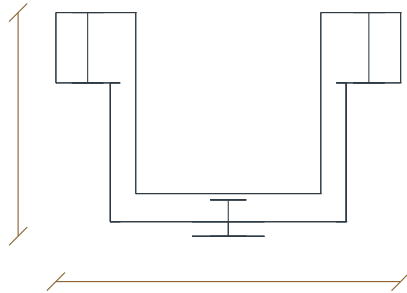
W 8.000 / H 38.000 / scale 1.2000  
CAD lower-left XY: 196.000, -19.000

XZ / positive axes right and up



W 8.000 / H 24.644 / scale 1.2000  
CAD lower-left XZ: 196.000, -24.894

YZ / positive axes right and up



W 38.000 / H 24.644 / scale 1.2000  
CAD lower-left YZ: -19.000, -24.894

Conventional split8mm-long saddle around stock20x40 tube; cavity20.4x40.4, seam0.5. Two ordinary clamp bolts; no stock-tube drilling or cable passage blockage. Tightening/friction requires physical fit.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

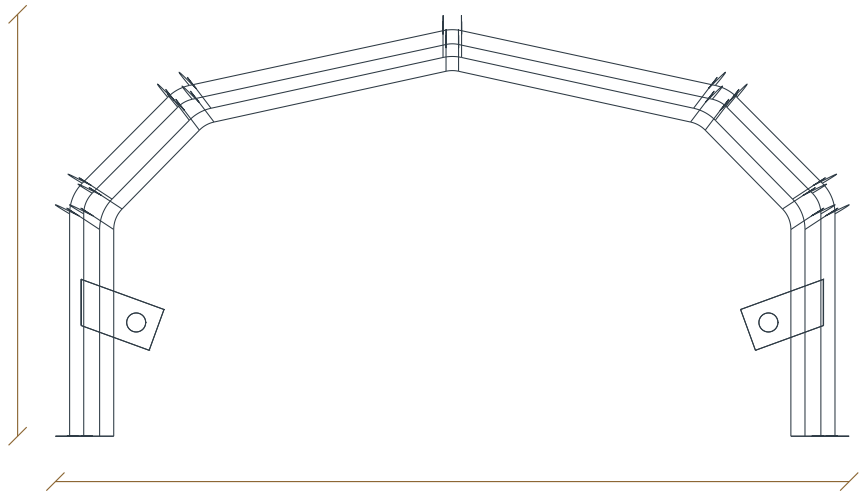
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C03-root-cowl-a

Frame J1.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

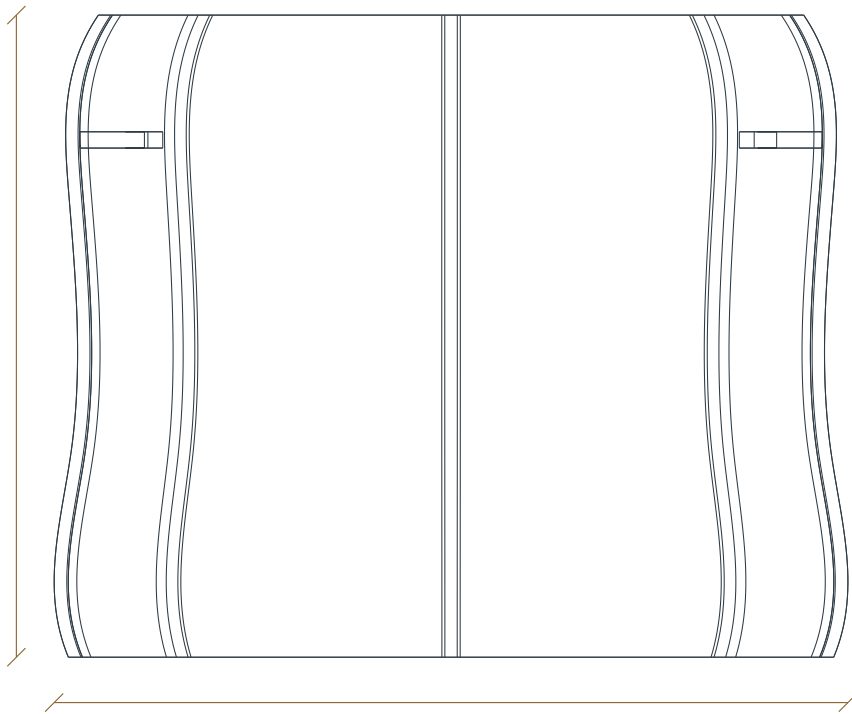
XY / positive axes right and up



W 146.290 / H 77.726 / scale 0.7178

CAD lower-left XY: -73.145, 0.250

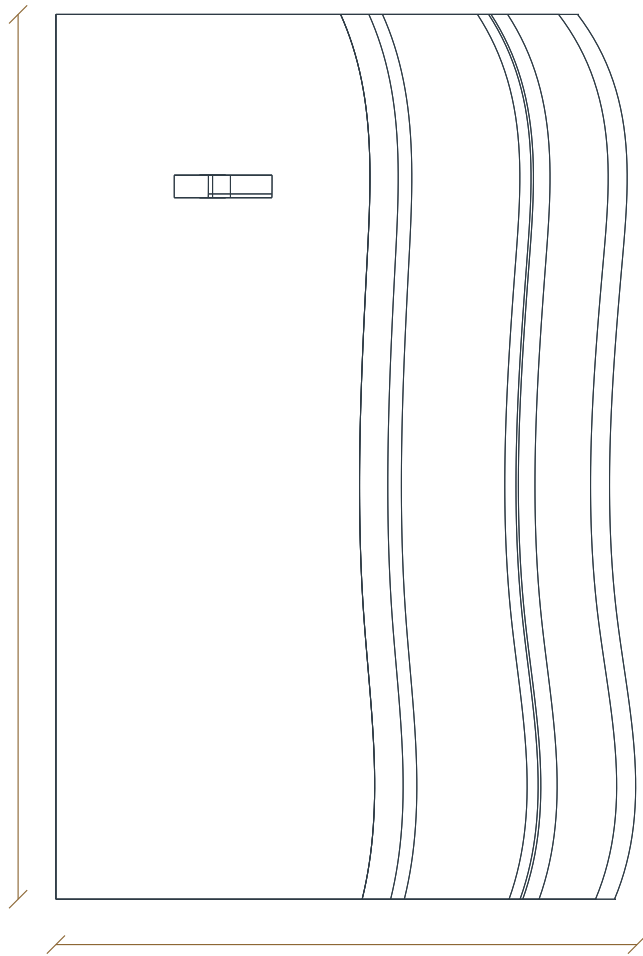
XZ / positive axes right and up



W 146.290 / H 118.300 / scale 0.7178

CAD lower-left XZ: -73.145, -89.800

YZ / positive axes right and up



W 77.726 / H 118.300 / scale 0.9890

CAD lower-left YZ: 0.250, -89.800

Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

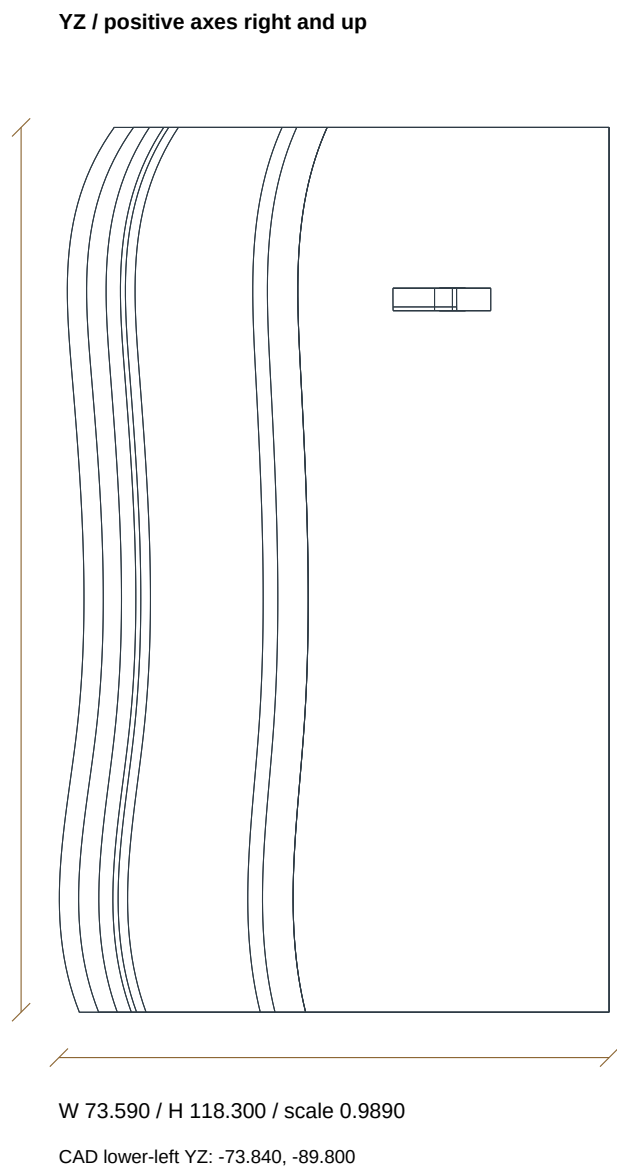
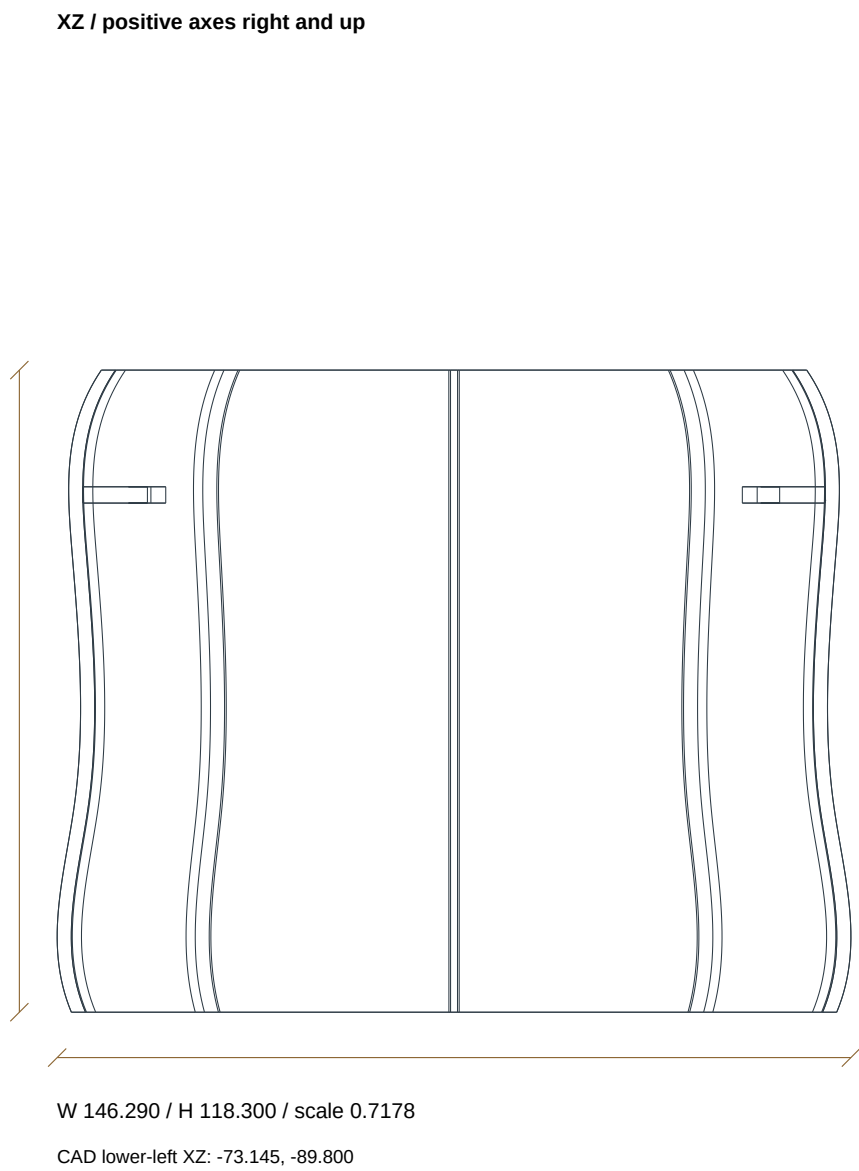
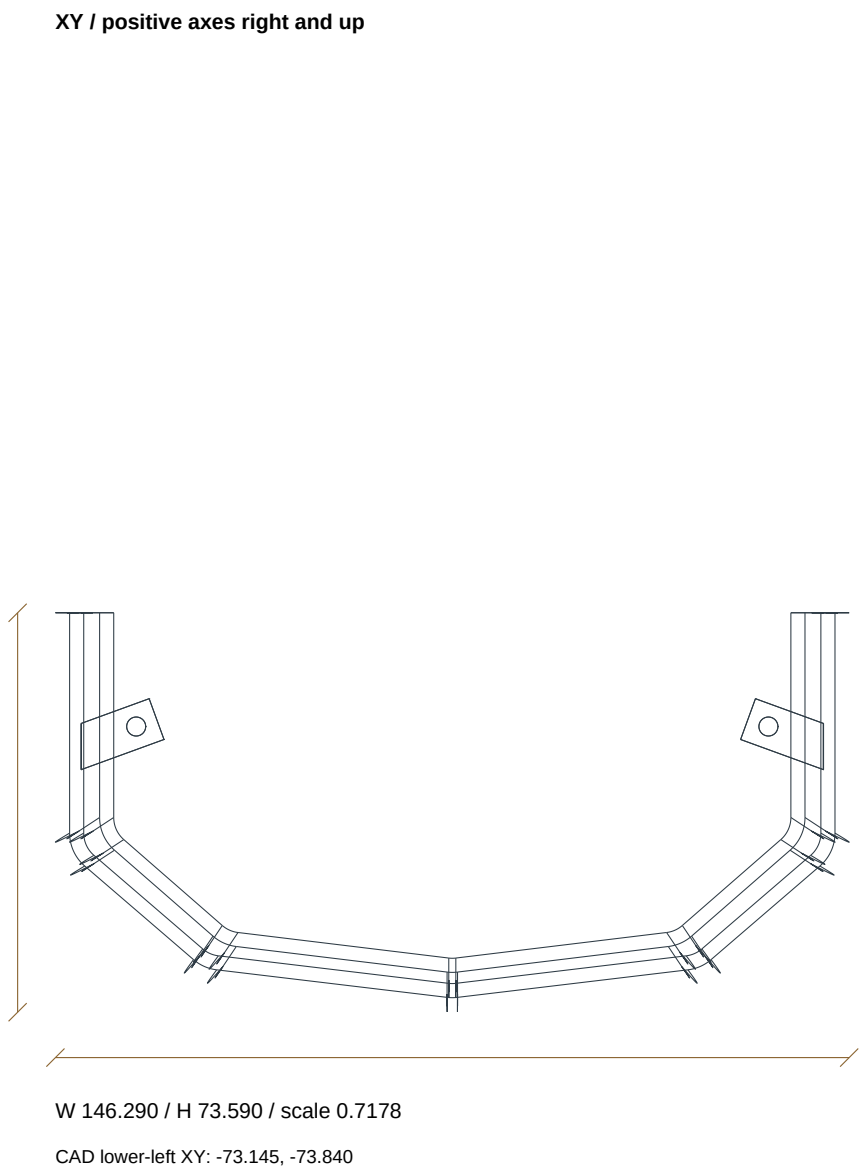
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C03-root-cowl-b

Frame J1.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

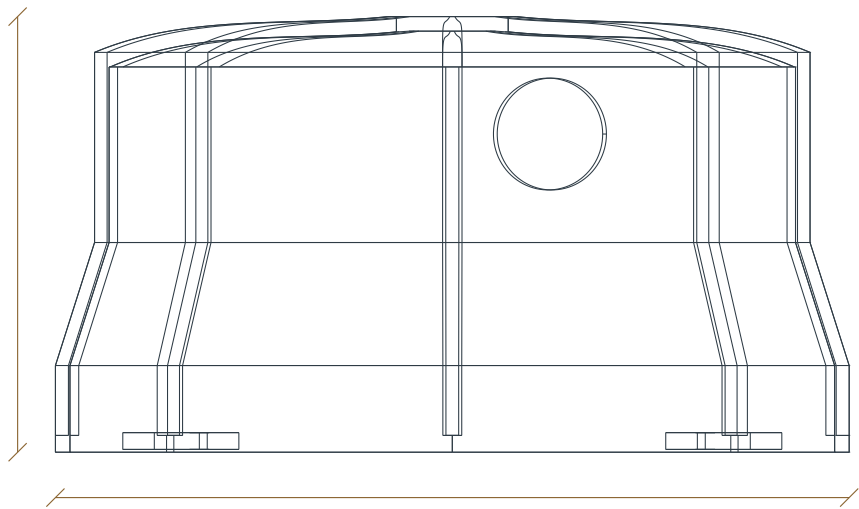
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C04-J2-cowl-a

Frame J2.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

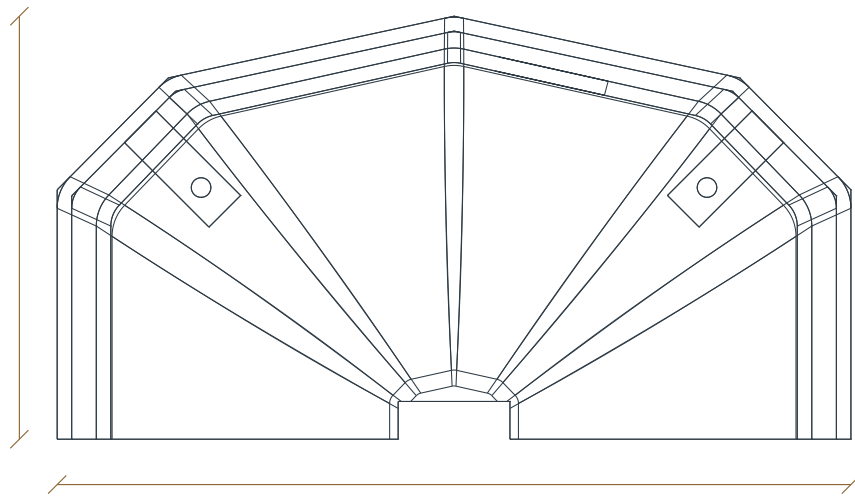
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



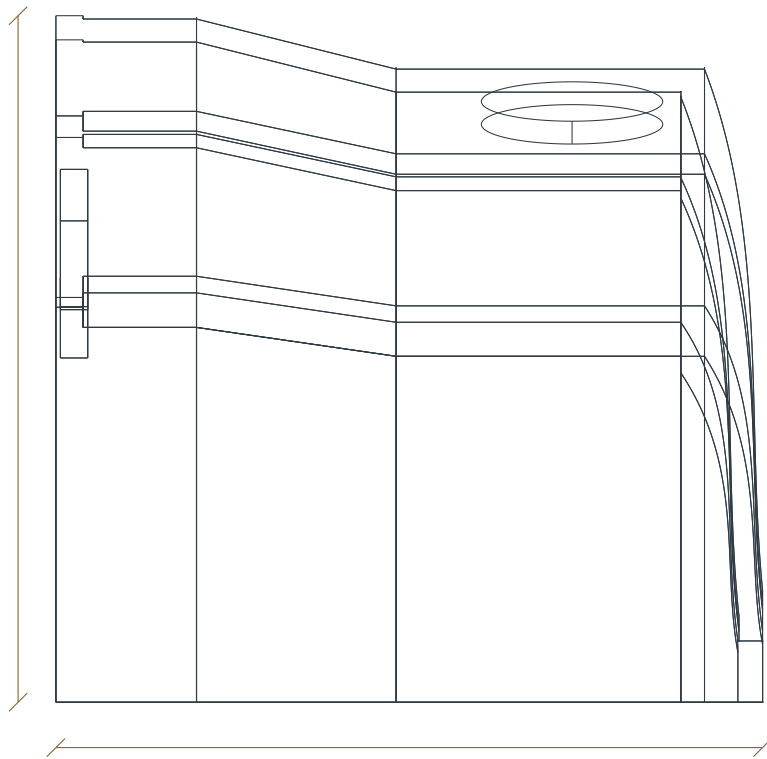
W 142.000 / H 77.900 / scale 0.7394  
CAD lower-left XY: -71.000, 56.500

XZ / positive axes right and up



W 142.000 / H 75.650 / scale 0.7394  
CAD lower-left XZ: -71.000, 0.250

YZ / positive axes right and up



W 77.900 / H 75.650 / scale 1.2000  
CAD lower-left YZ: 56.500, 0.250

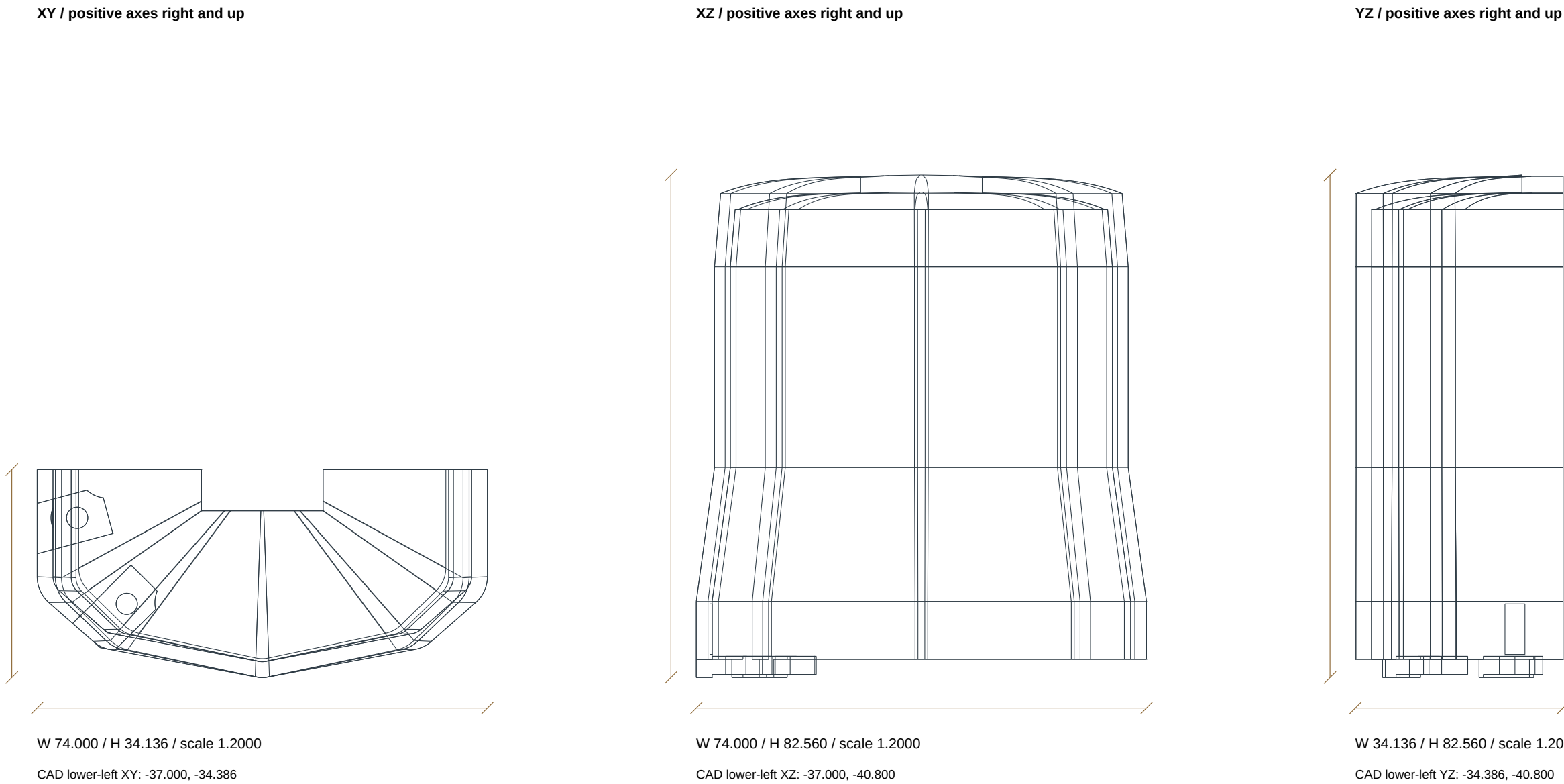
Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C05-J6-cowl-a

Frame J6.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



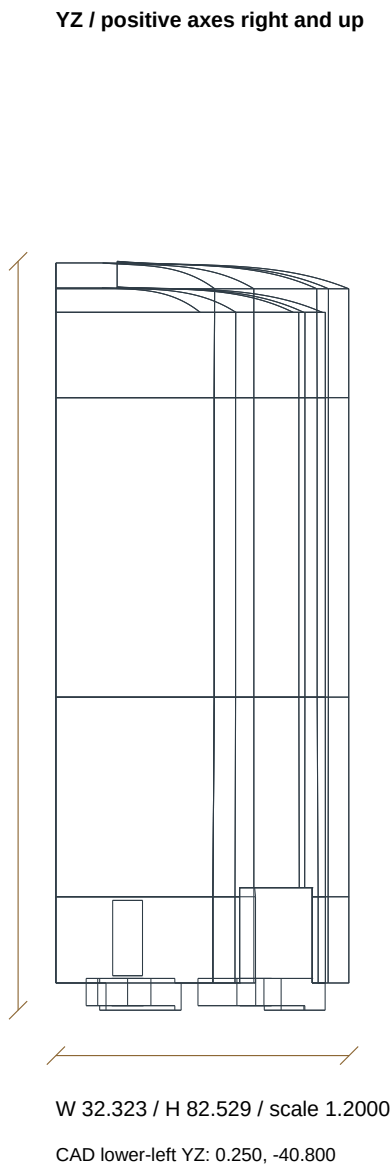
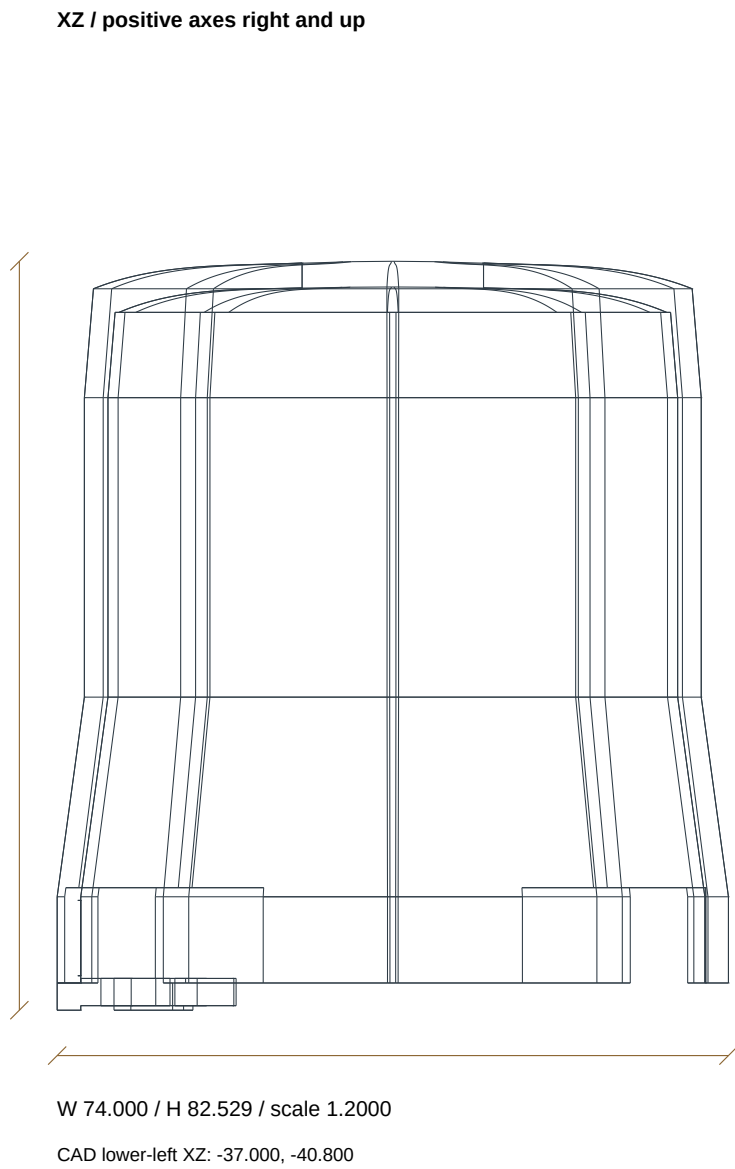
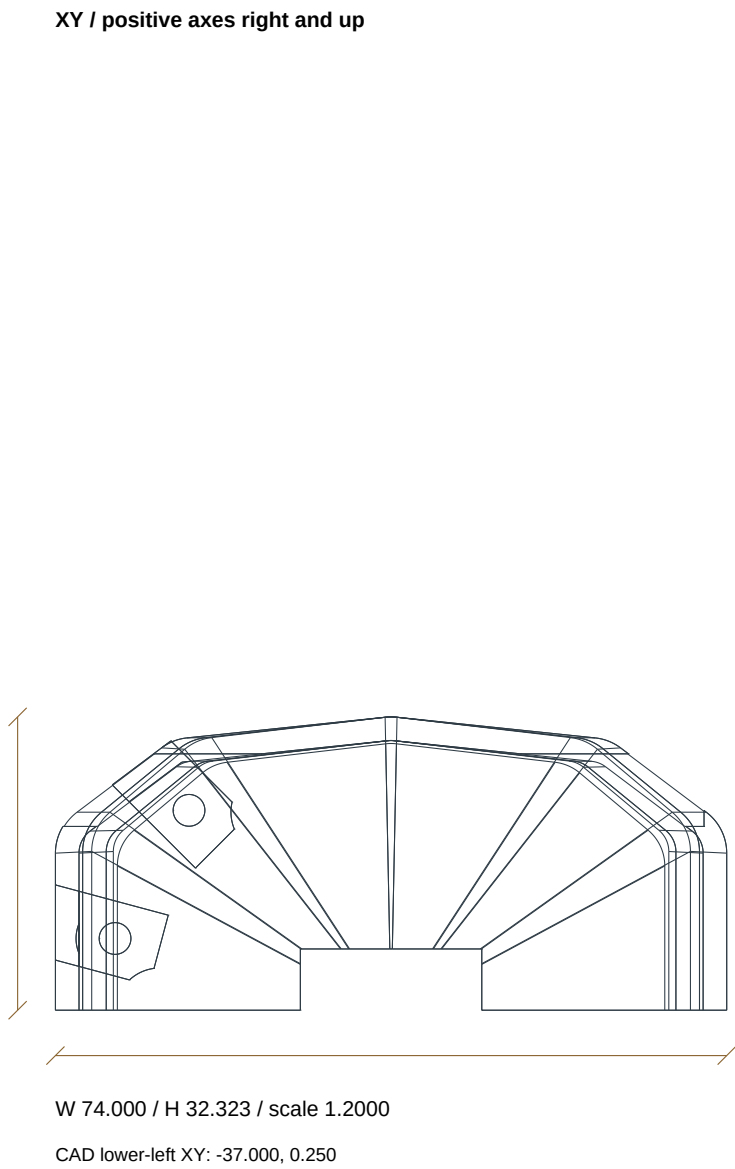
Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C05-J6-cowl-b

Frame J6.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



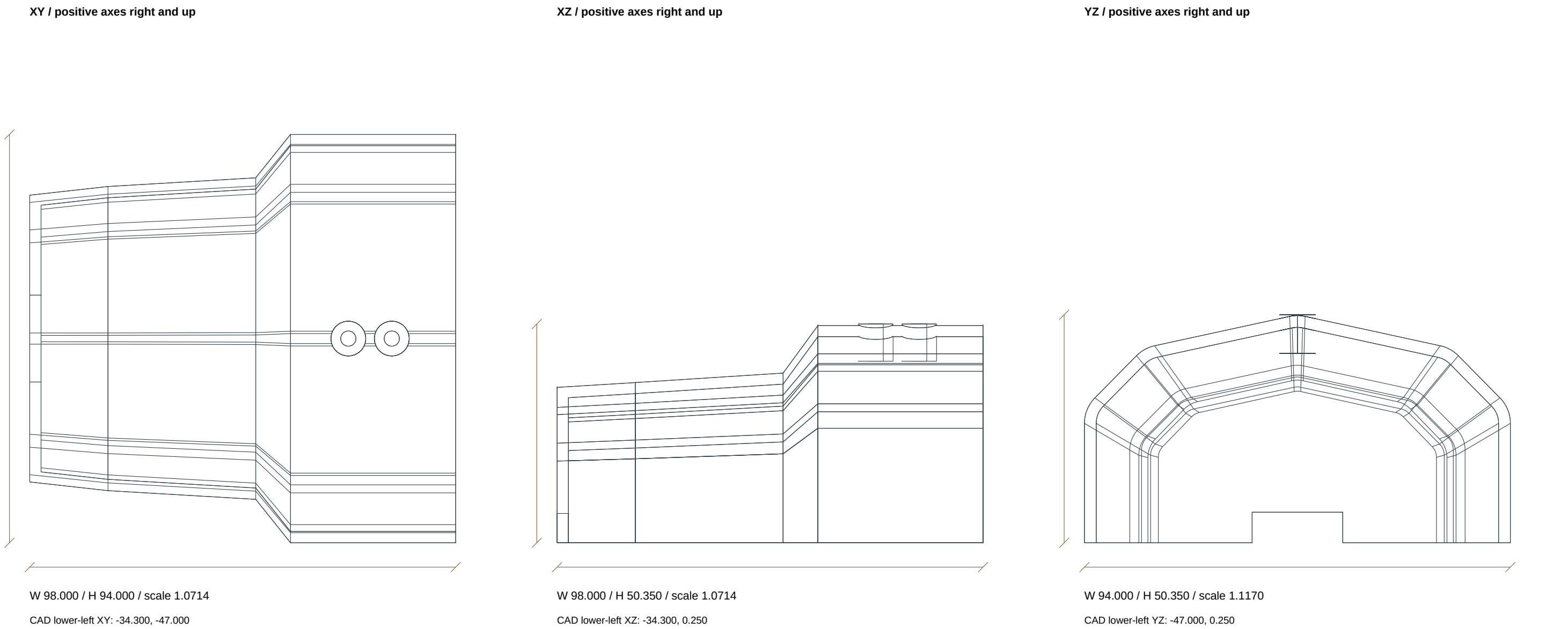
Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C05-J7-cowl-a

Frame J7.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



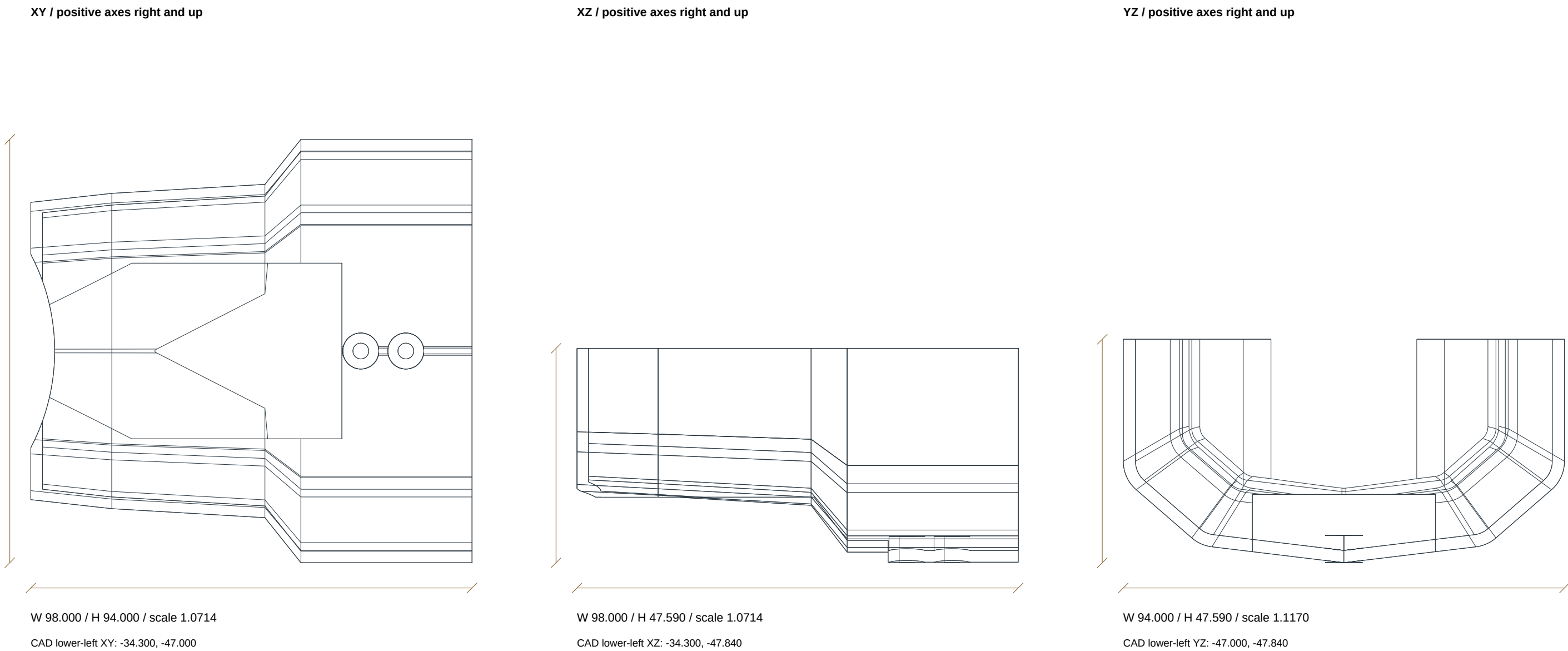
Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C05-J7-cowl-b

Frame J7.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

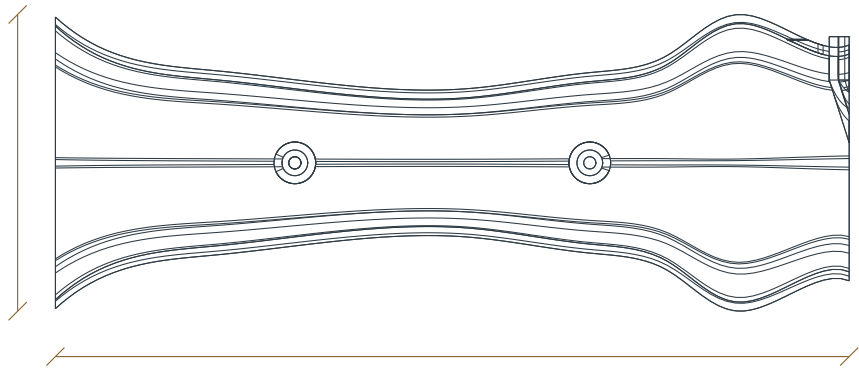
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C06-upper-dorsal

Frame J3.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

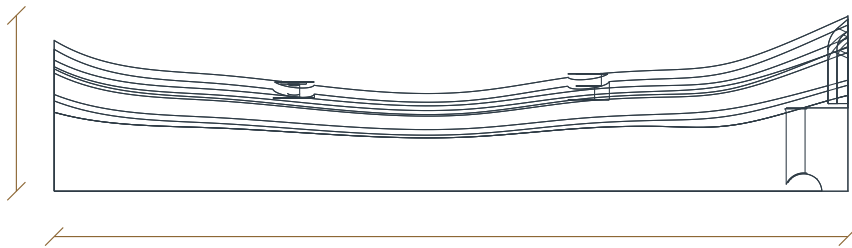
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



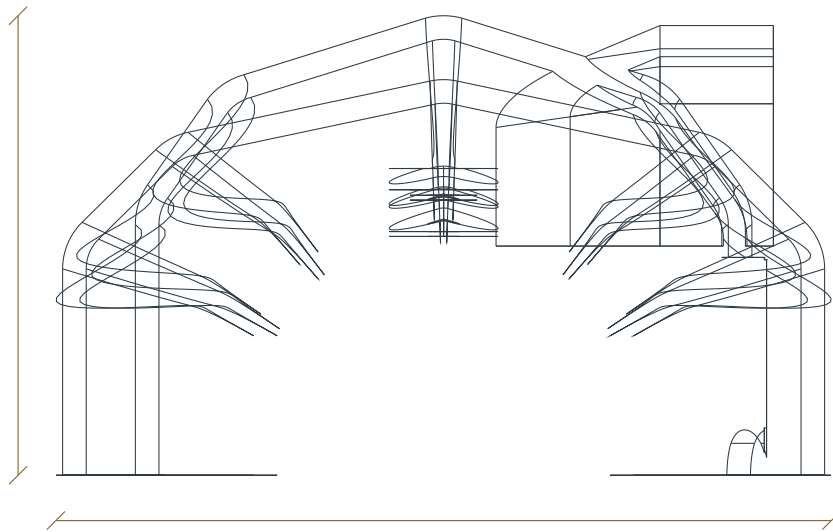
W 228.750 / H 85.482 / scale 0.4590  
CAD lower-left XY: 46.000, -42.741

XZ / positive axes right and up



W 228.750 / H 50.632 / scale 0.4590  
CAD lower-left XZ: 46.000, 0.250

YZ / positive axes right and up



W 85.482 / H 50.632 / scale 1.2000  
CAD lower-left YZ: -42.741, 0.250

Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

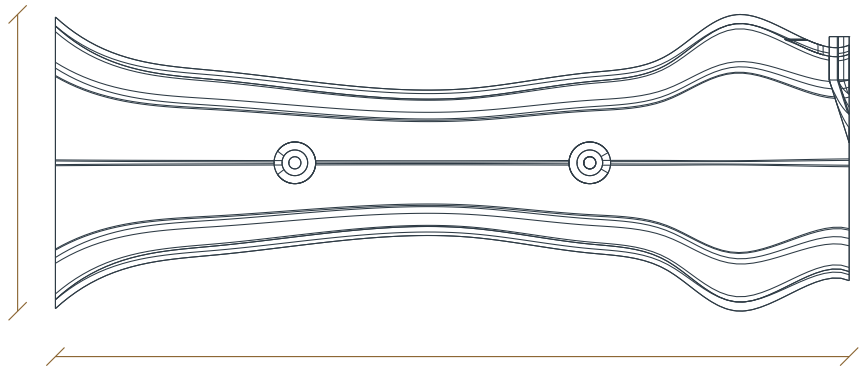
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A17-C06-upper-ventral

Frame J3.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

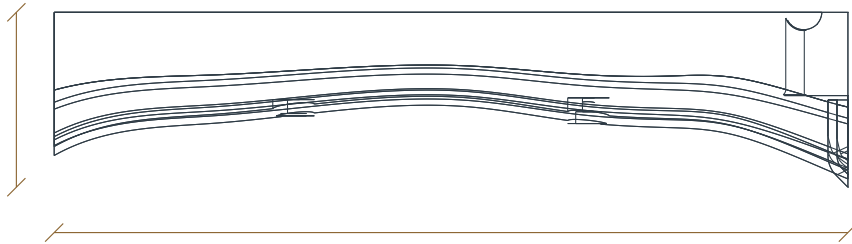
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



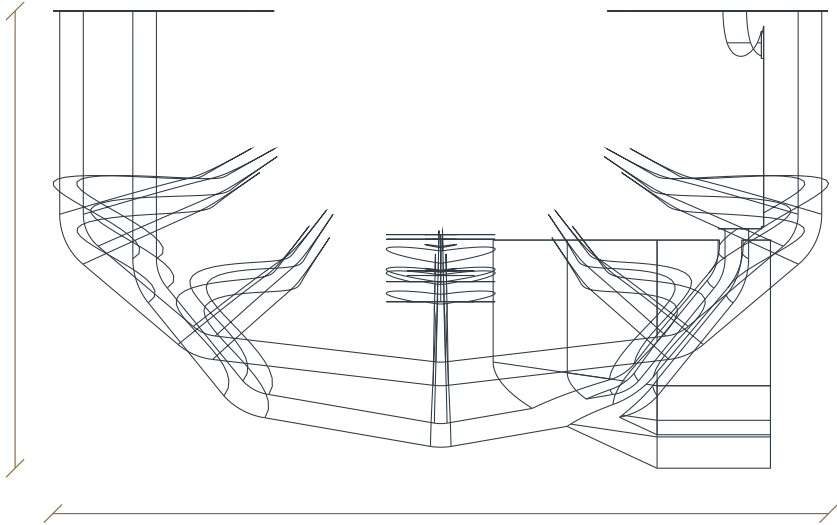
W 228.750 / H 85.504 / scale 0.4590  
CAD lower-left XY: 46.000, -42.752

XZ / positive axes right and up



W 228.750 / H 50.388 / scale 0.4590  
CAD lower-left XZ: 46.000, -50.638

YZ / positive axes right and up



W 85.504 / H 50.388 / scale 1.2000  
CAD lower-left YZ: -42.752, -50.638

Rounded CAD section with unchanged C03-C06 mounting features and reliefs. Nominal section2.6mm; global wall, physical fit, mass and motion require fresh validation. Appearance development candidate, not production. J4 fixed socket axial clearance cuts at +/-0.25mm, no change to mounting datums.  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

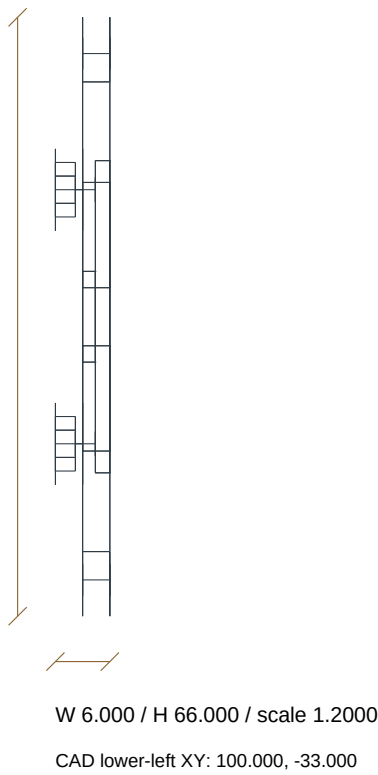
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A18-IF09-101-contact-insert

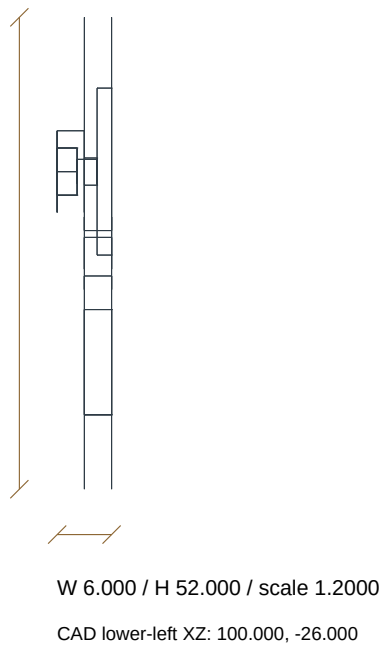
Frame J7.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

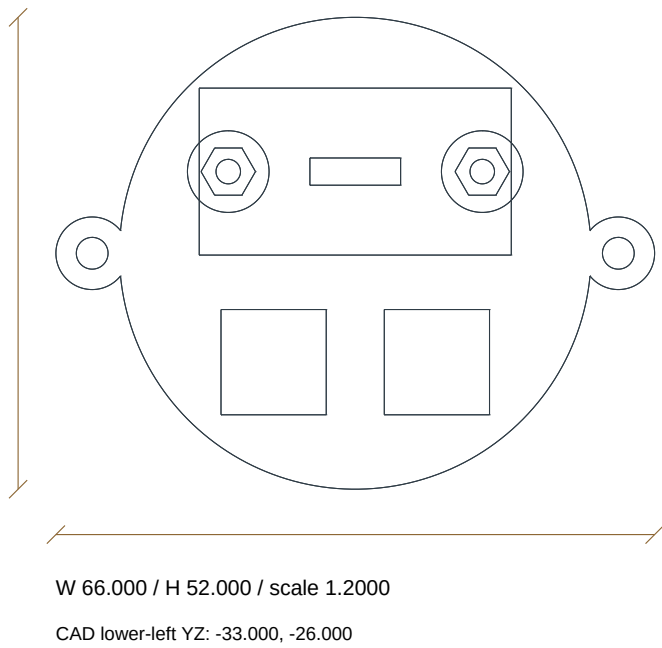
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



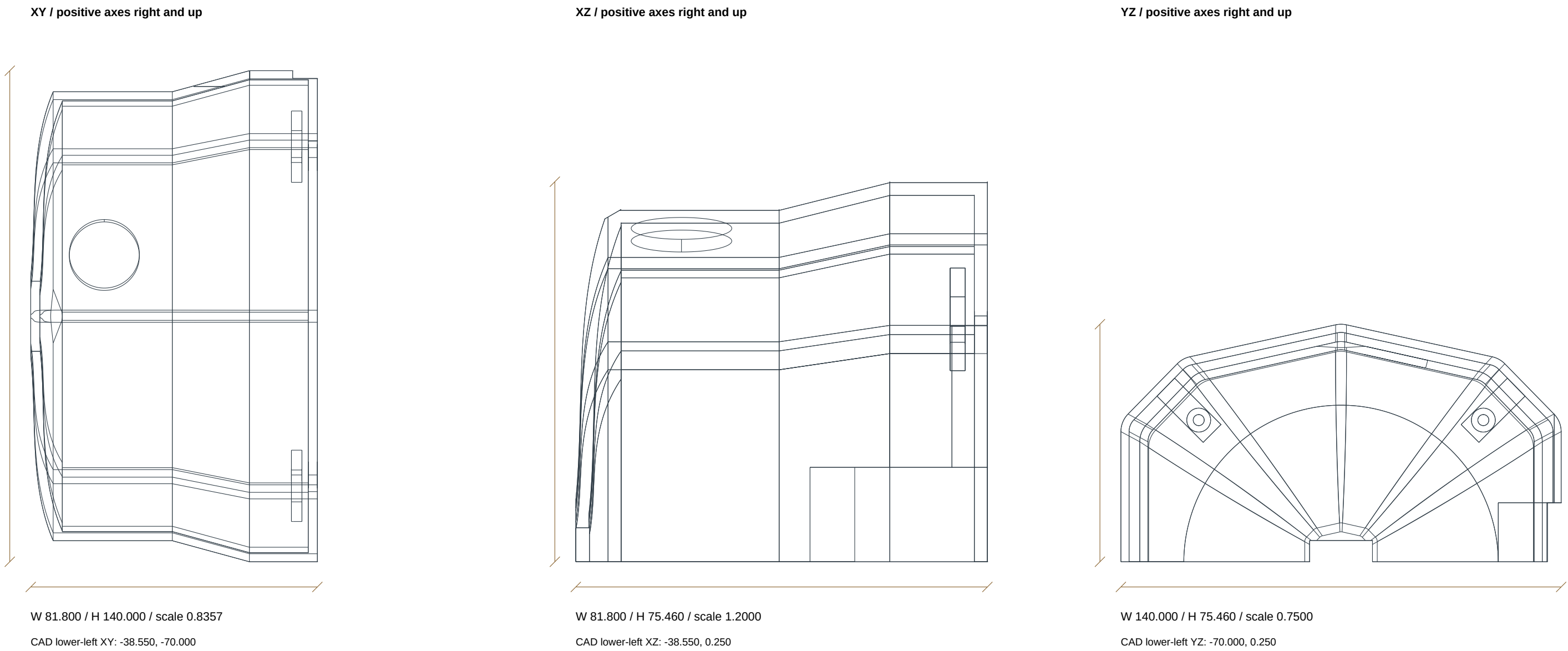
PETG/PA12 supported unpowered fit only  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A18-C04-J3-cowl-a

Frame J3.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



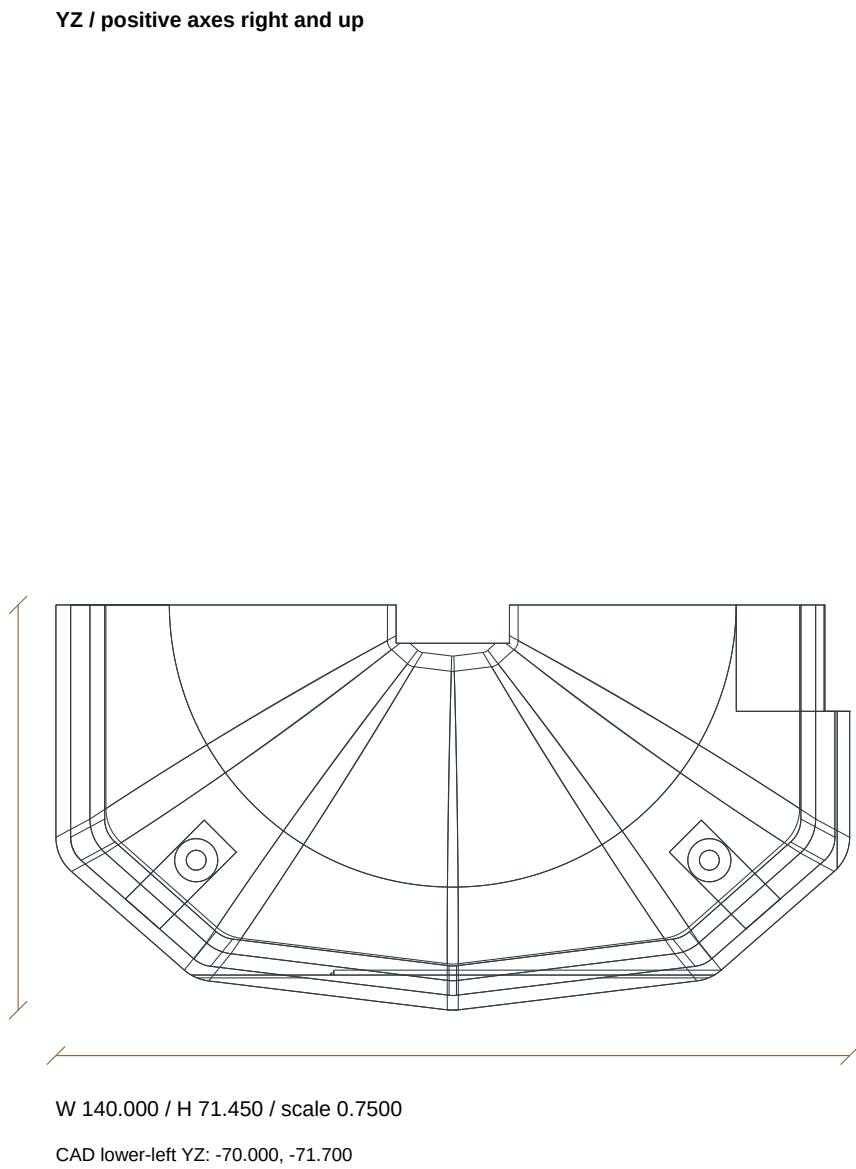
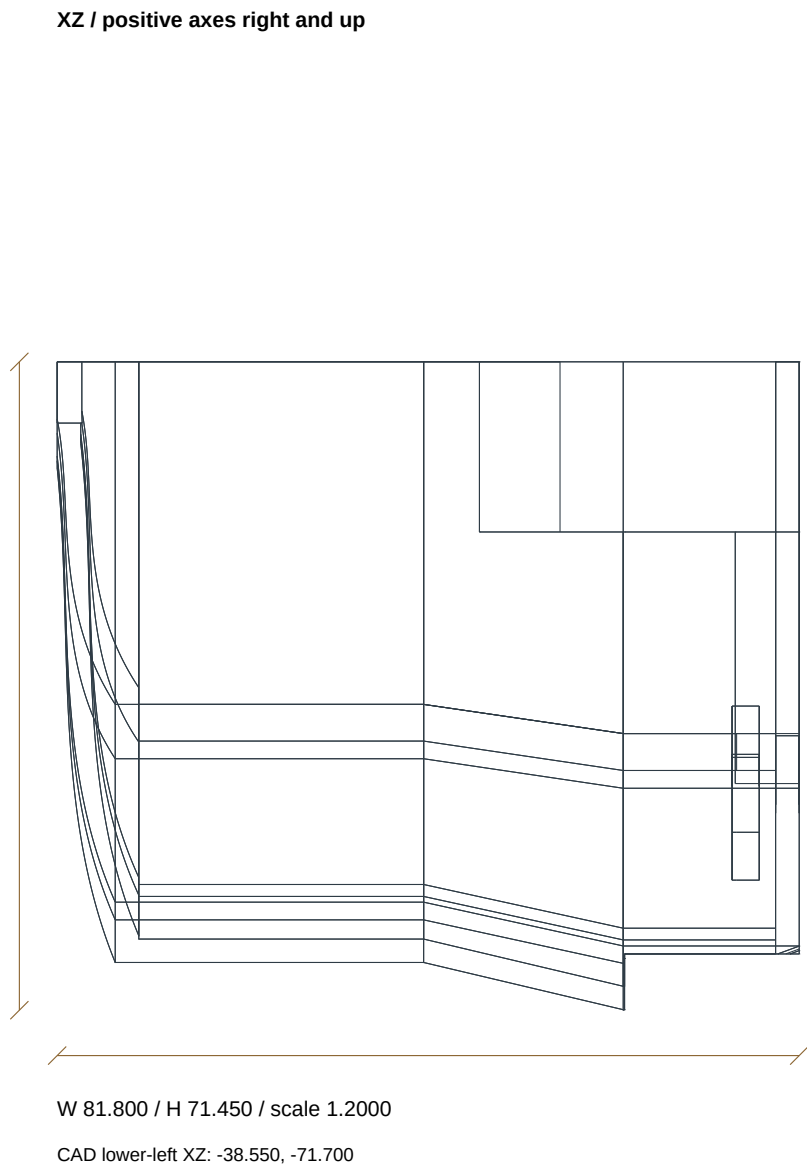
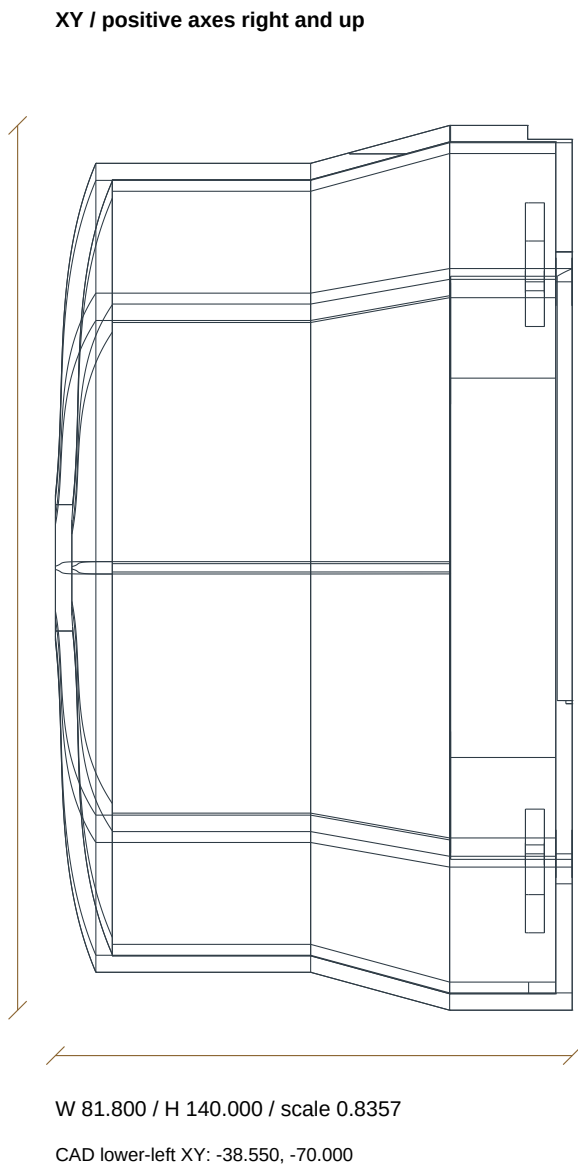
PETG/PA12 supported unpowered fit  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A18-C04-J3-cowl-b

Frame J3.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



PETG/PA12 supported unpowered fit  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

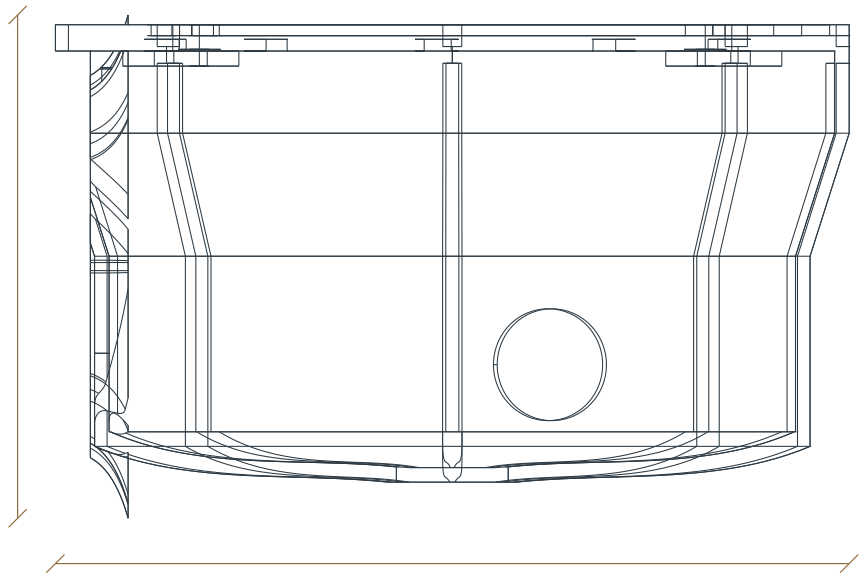
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A18-C04-J4-cowl-a

Frame J3.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

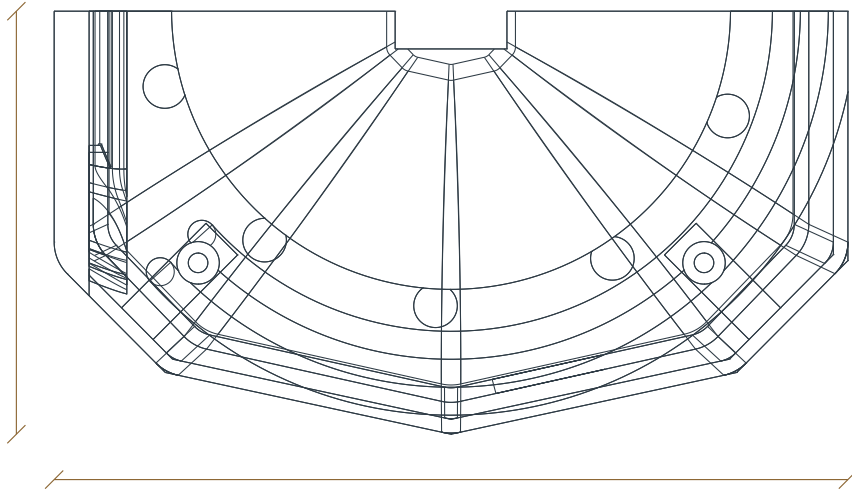
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



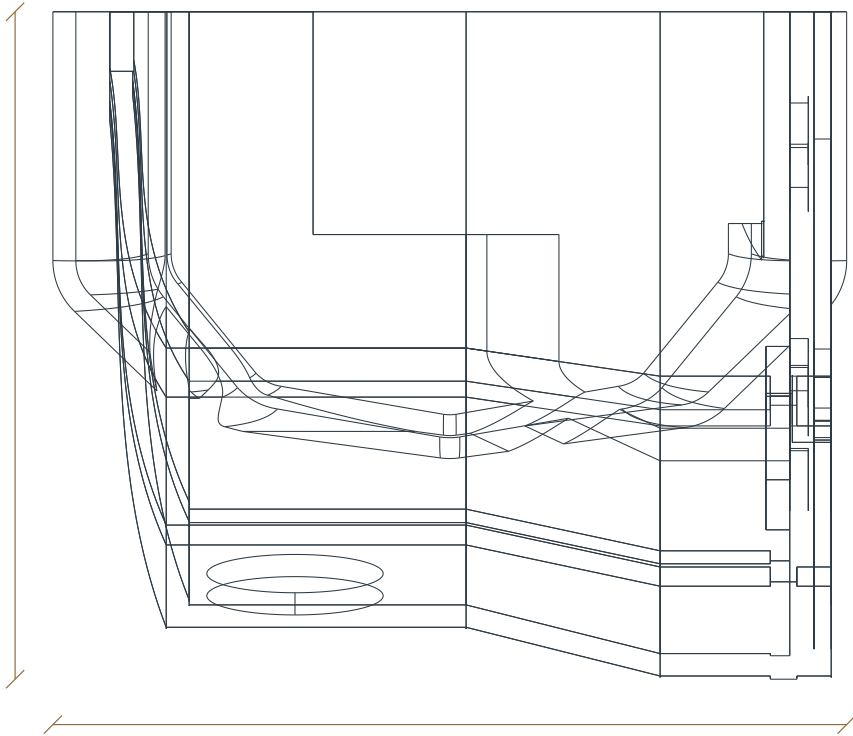
W 142.000 / H 90.000 / scale 0.7394  
CAD lower-left XY: 269.000, -45.000

XZ / positive axes right and up



W 142.000 / H 75.650 / scale 0.7394  
CAD lower-left XZ: 269.000, -75.900

YZ / positive axes right and up



W 90.000 / H 75.650 / scale 1.1667  
CAD lower-left YZ: -45.000, -75.900

PETG/PA12 supported unpowered fit  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

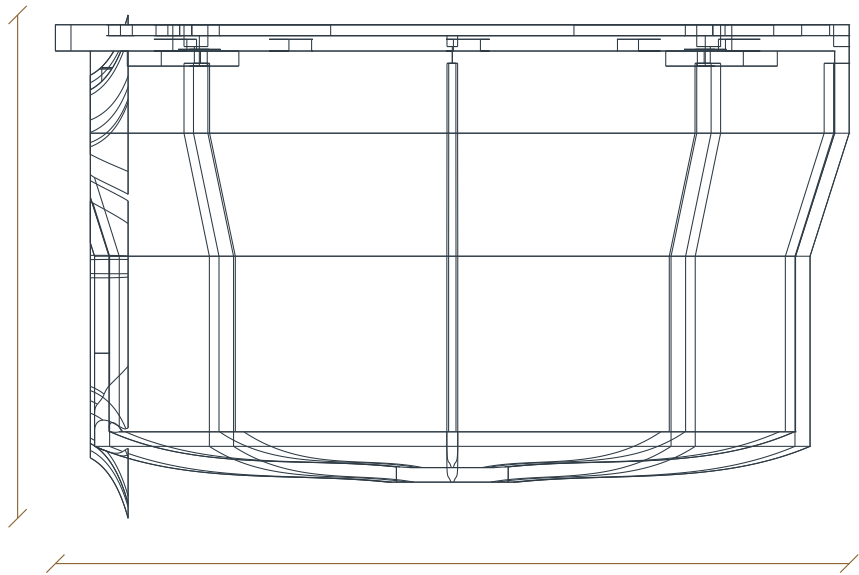
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A18-C04-J4-cowl-b

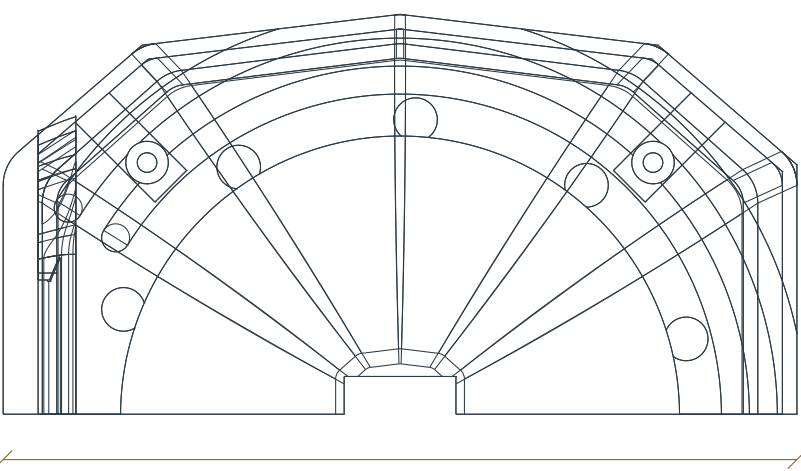
Frame J3.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

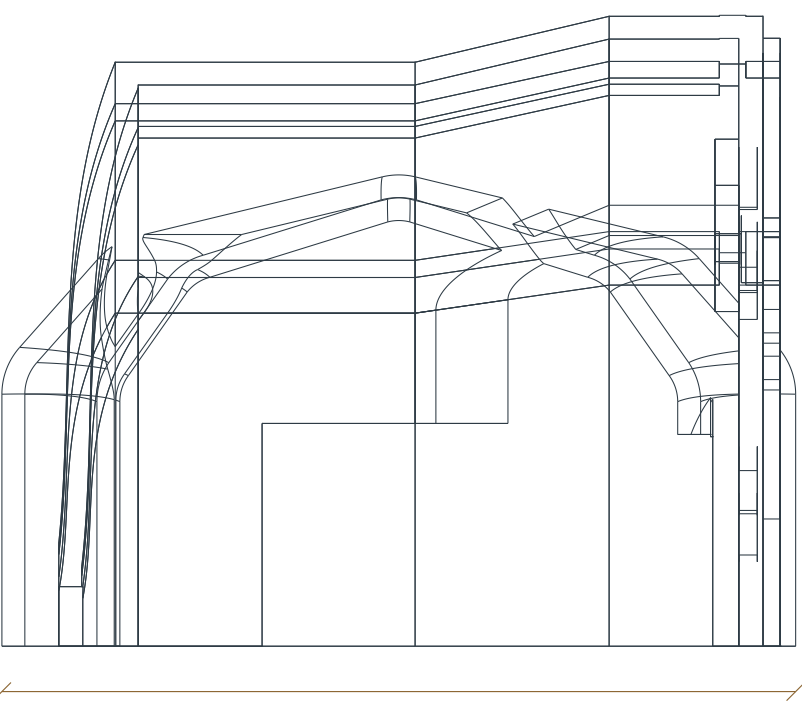
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



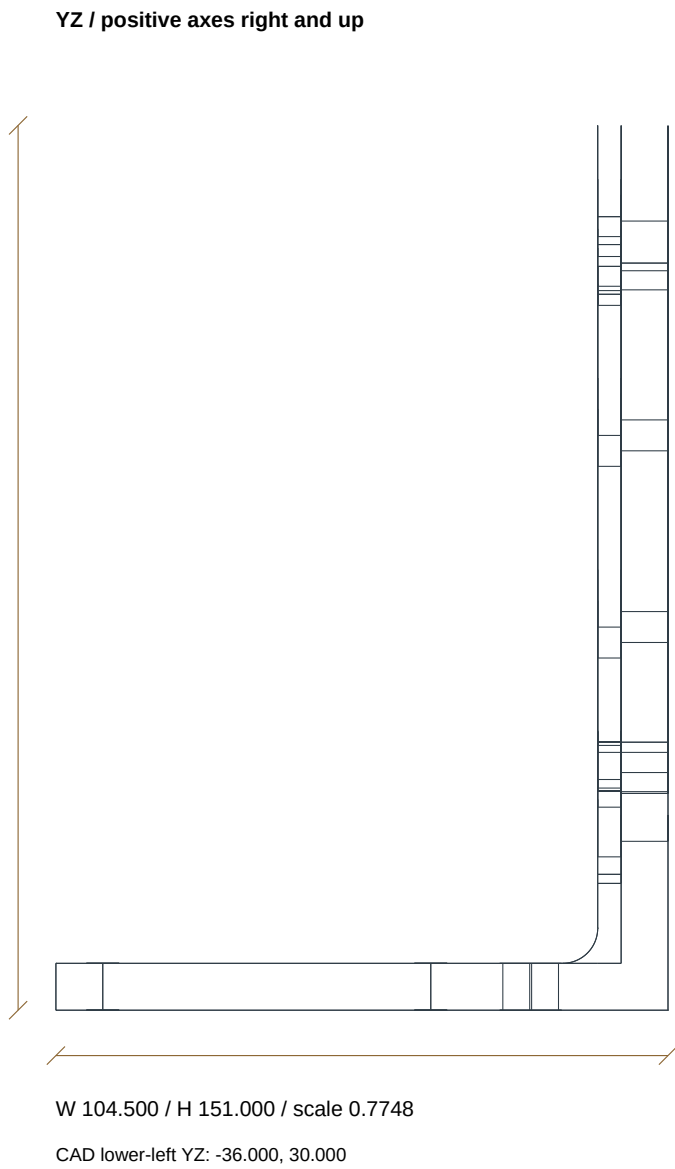
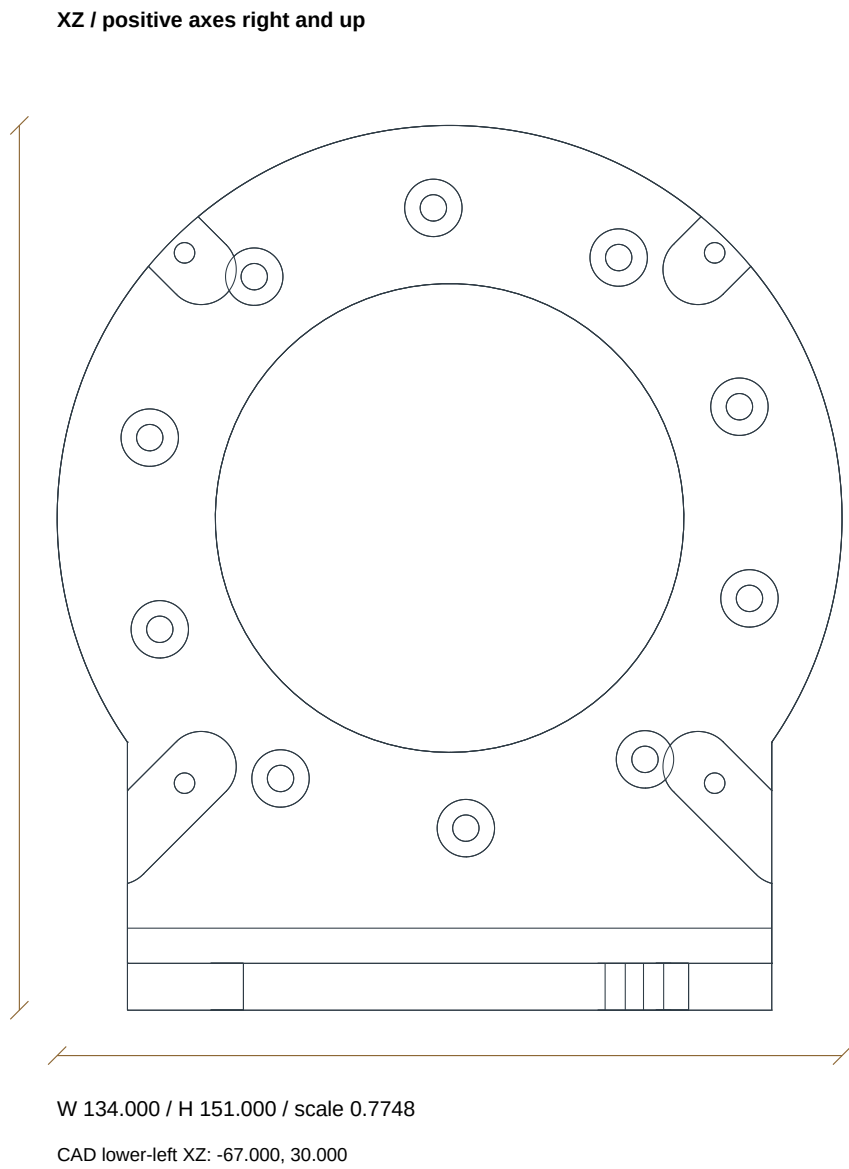
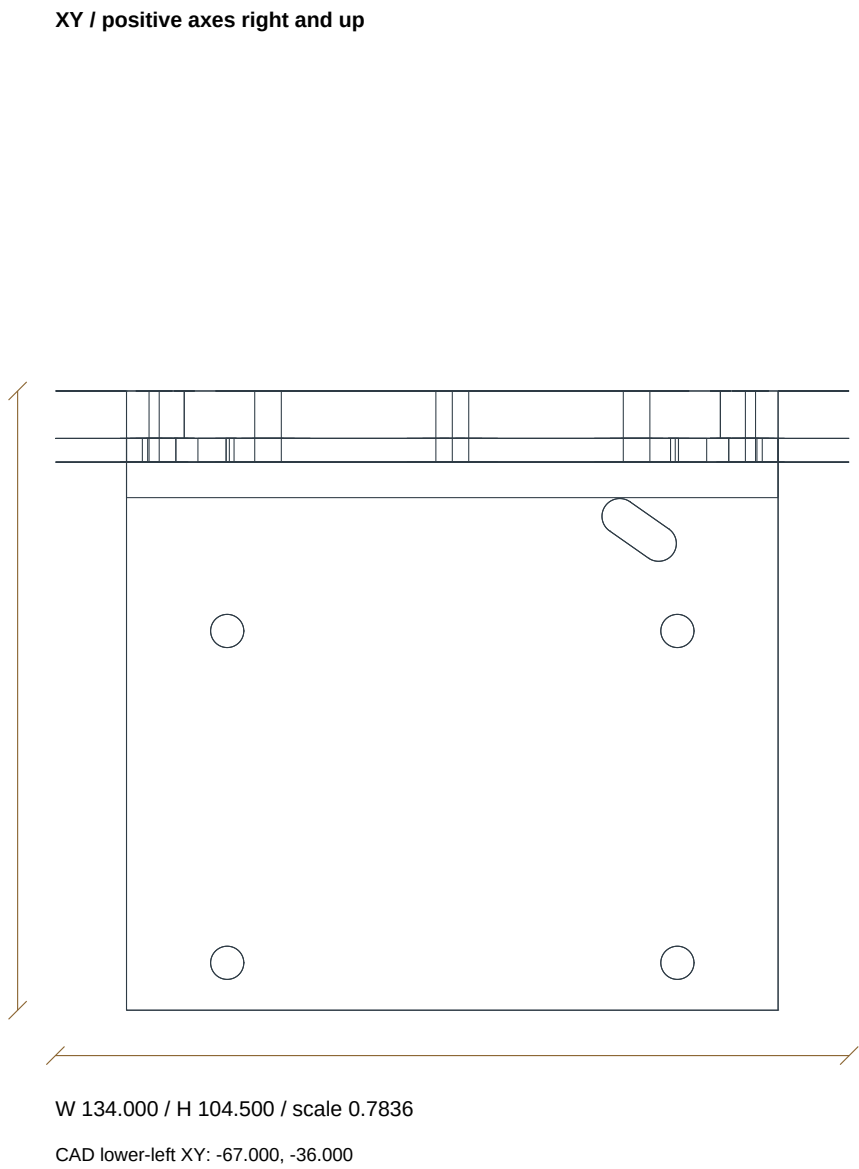
PETG/PA12 supported unpowered fit  
Retained nominal source geometry. Critical pilots, bearing seats and nut fit require physical measurement.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-S22-shoulder-foot-12mm-R6

Frame J1.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Shoulder axis X0/Z114; motor mating planeY68.5 in J1.rotor. Face12, OD134/ID80.  
Foot110x104.5x8; bottomZ30. FourD5.6 centresX+/-38,Y+/-28. Inner rootR6.  
Native tenD4.5 motor holes; frontD9.8 counterboresdepth4 retain8mm grip.  
Original M4x12 + washer0.8 => insertion3.2. Keep actual mounting centres.  
Four R64 cover seats45/135/225/315deg retain8mm underneath ordinary open radial pockets.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

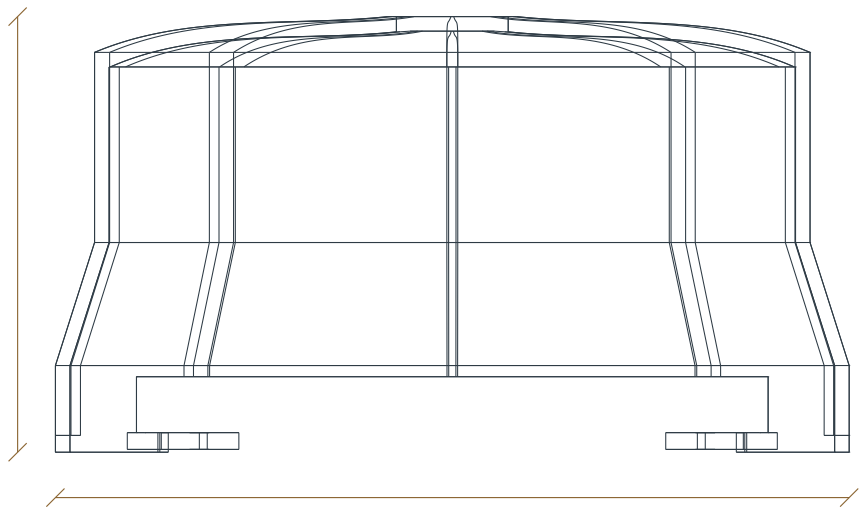
STEP SHA256 77e0c1a5ea389a546f54ad87... | Source assembly 4e76c049091d411b...

A19-C22-J2-cowl-b

Frame J2.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

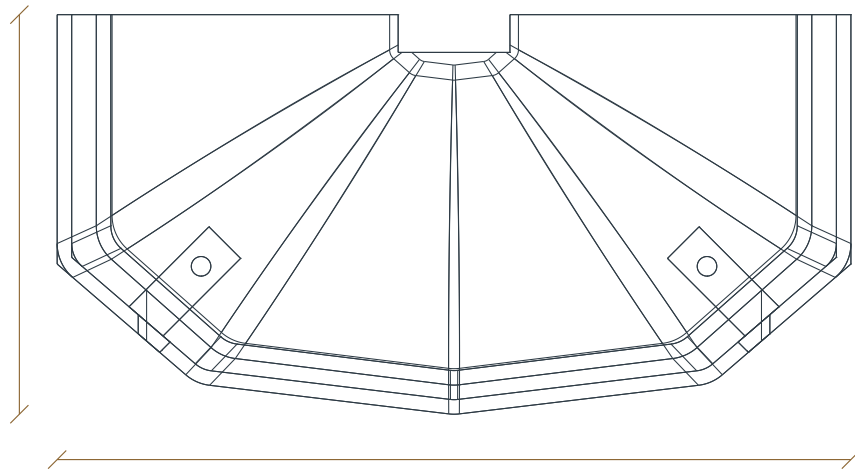
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



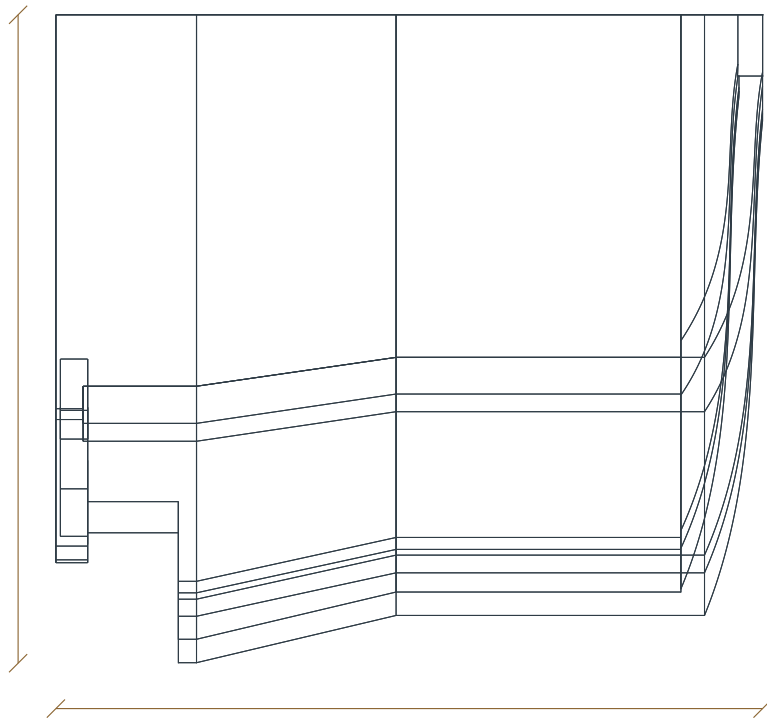
W 142.000 / H 77.900 / scale 0.7394  
CAD lower-left XY: -71.000, 56.500

XZ / positive axes right and up



W 142.000 / H 71.453 / scale 0.7394  
CAD lower-left XZ: -71.000, -71.703

YZ / positive axes right and up



W 77.900 / H 71.453 / scale 1.2000  
CAD lower-left YZ: 56.500, -71.703

Lower shoulder split cover locally relieved for current12mm root bracket.  
Original fixing stations and upper cowl retained. Never mix with original8mm shoulder plate.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

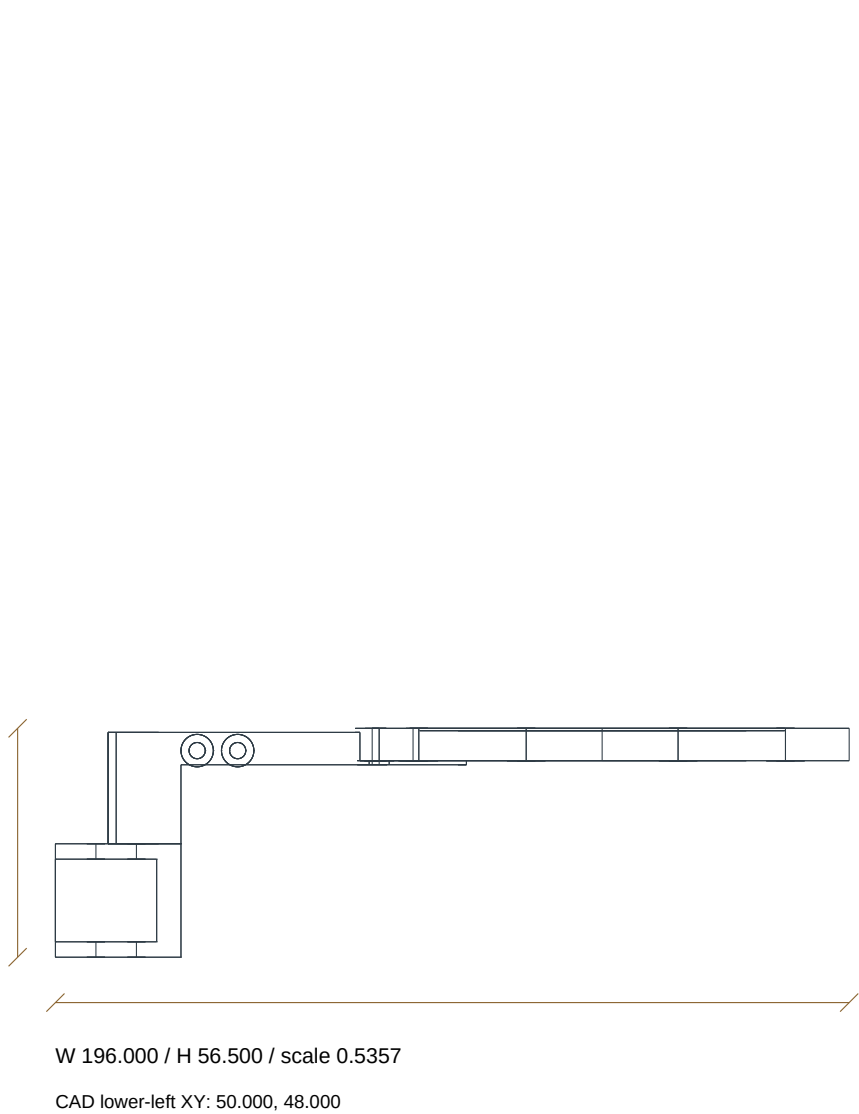
STEP SHA256 3469a012d72c5ff132996a4d... | Source assembly 4e76c049091d411b...

A19-S19-fore-spine-RS03-ring

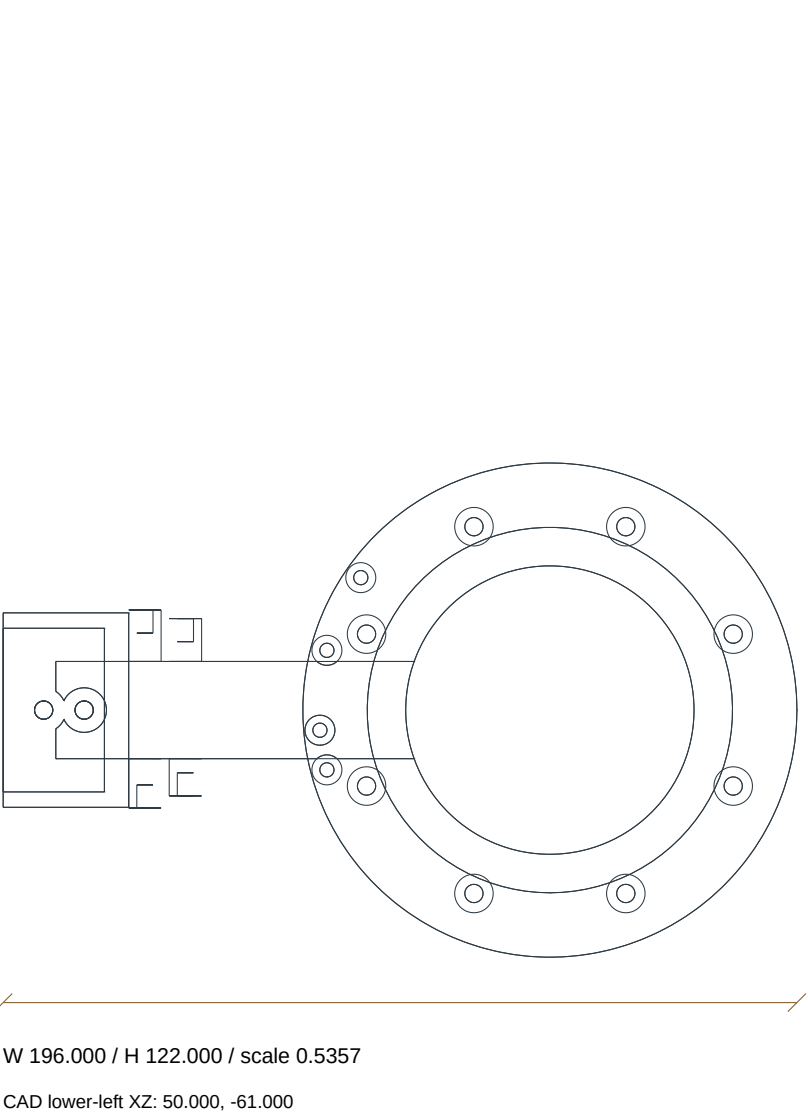
Frame J4.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

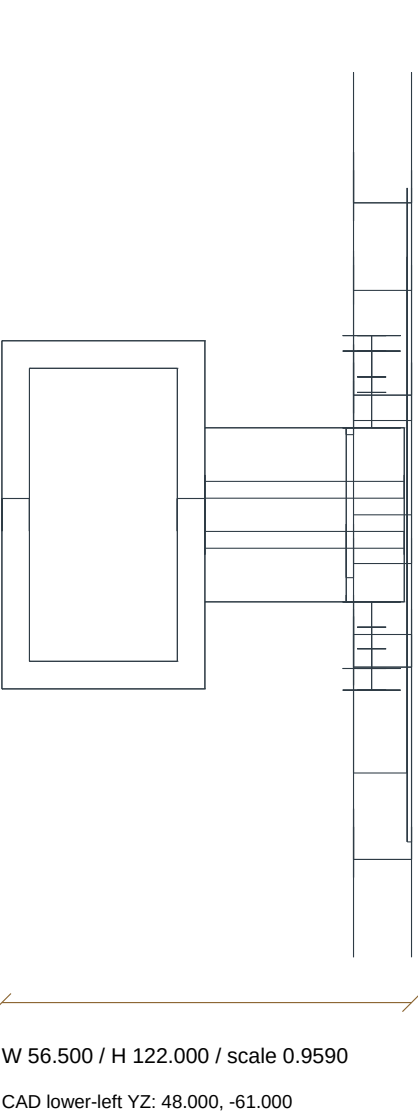
XY / positive axes right and up



XZ / positive axes right and up



YZ / positive axes right and up



Motor face in J4.rotor: Y104.5; native J5 axis X185 / Z0. 8mm plate, OD122 / central ID71.2.  
Raised native seal relief: OD90.2, depth0.6. Do not move motor mating plane.  
8xD4.5 motor holes; actual coordinates in current-native-hole-table.csv. M4x14 + washer0.8 => insertion5.2.  
Four cover seats R57 at145/165/185/195deg; D3.5 through drill, D7.4 flat clearance.  
Retain stock20x40x2 socket and original fore-cover seats. Extra socket drilling is intentional.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

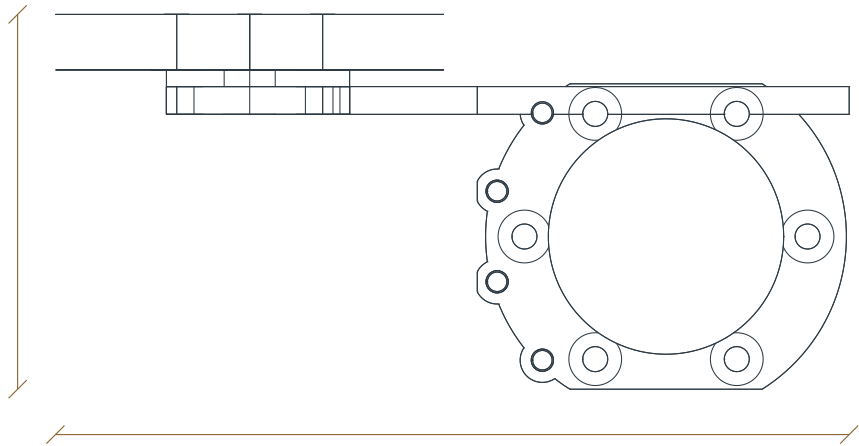
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-S19-RS03-to-J6-L

Frame J5.rotor | printed\_structure | mm | NOMINAL SUPPORTED FIT ISSUE

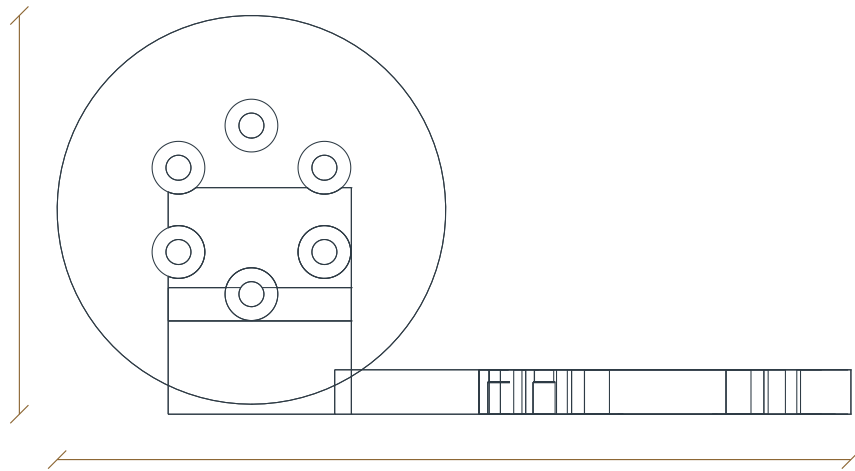
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



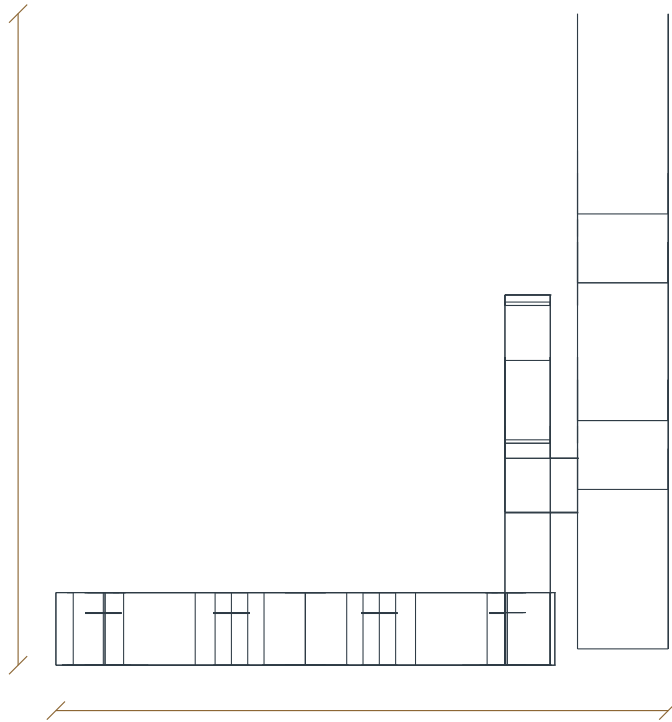
W 143.000 / H 67.500 / scale 0.7343  
CAD lower-left XY: -35.000, -27.500

XZ / positive axes right and up



W 143.000 / H 71.800 / scale 0.7343  
CAD lower-left XZ: -35.000, -36.800

YZ / positive axes right and up



W 67.500 / H 71.800 / scale 1.2000  
CAD lower-left YZ: -27.500, -36.800

Native J5 output plane Y40 in J5.rotor; OD70, plate10 thick. SixD4.5 holes.  
M4x16 + washer0.8 => insertion5.2; verify real native6mm depth and entry chamfer.  
J6 fixed interface centre X75 / Z-28.8; sixD4.5 holes on PCD51. J6 is actual RS10P, not RS02.  
Existing J6 cover seats shiftedX20 with the complete local heatset/hardware set.  
Use ordinary straight L webs; install output bolts before J6 blocks the shaft access.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

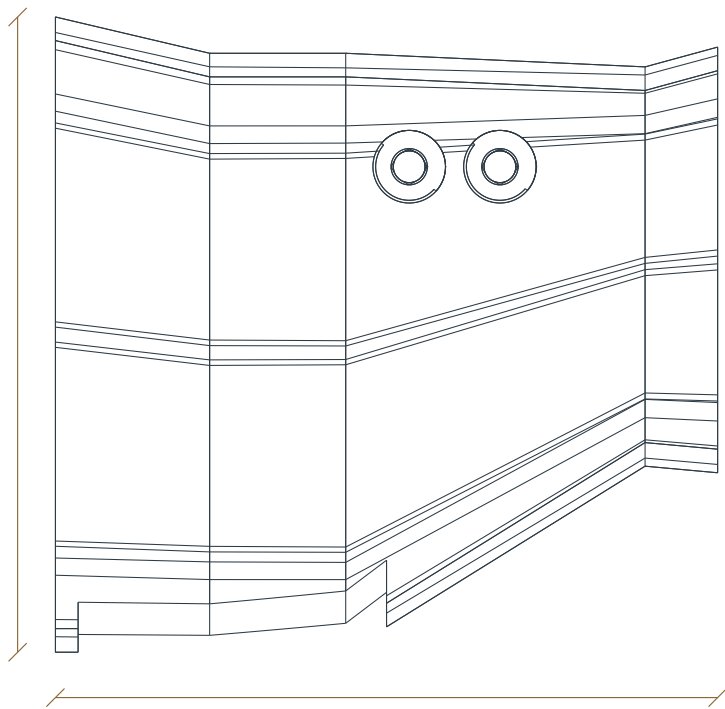
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-C19-fore-dorsal

Frame J4.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

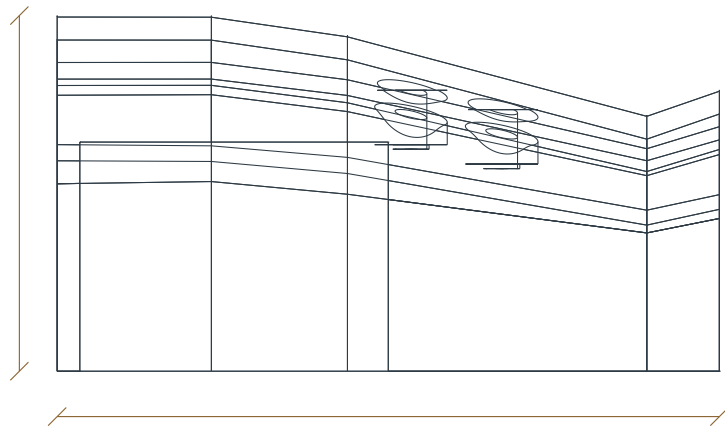
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



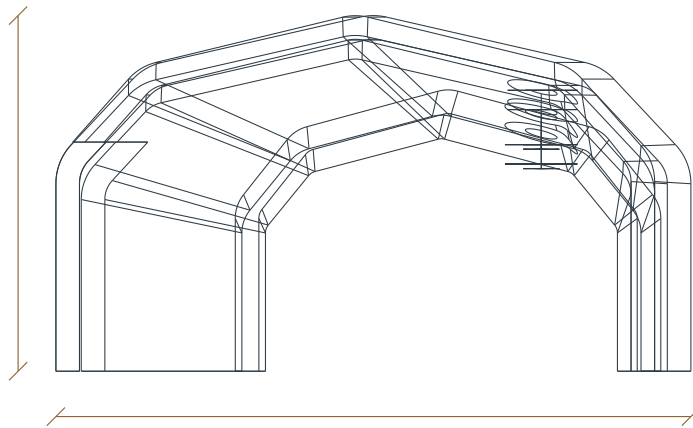
W 73.000 / H 70.000 / scale 1.2000  
CAD lower-left XY: 46.000, 45.500

XZ / positive axes right and up



W 73.000 / H 39.188 / scale 1.2000  
CAD lower-left XZ: 46.000, 0.250

YZ / positive axes right and up



W 70.000 / H 39.188 / scale 1.2000  
CAD lower-left YZ: 45.500, 0.250

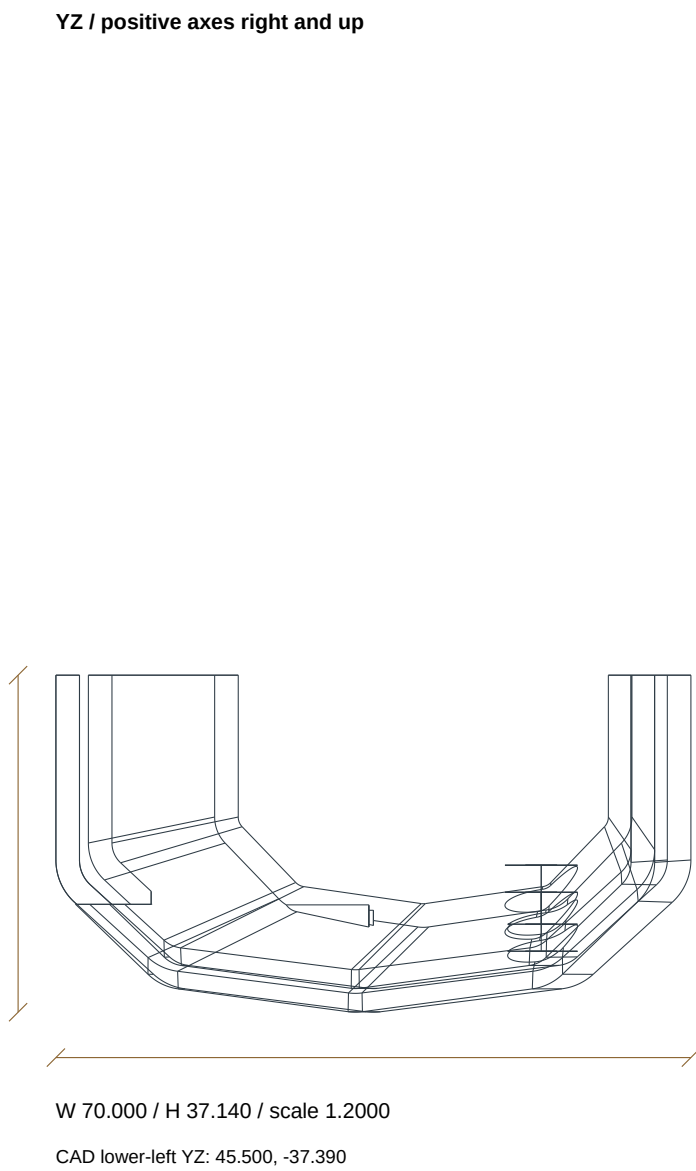
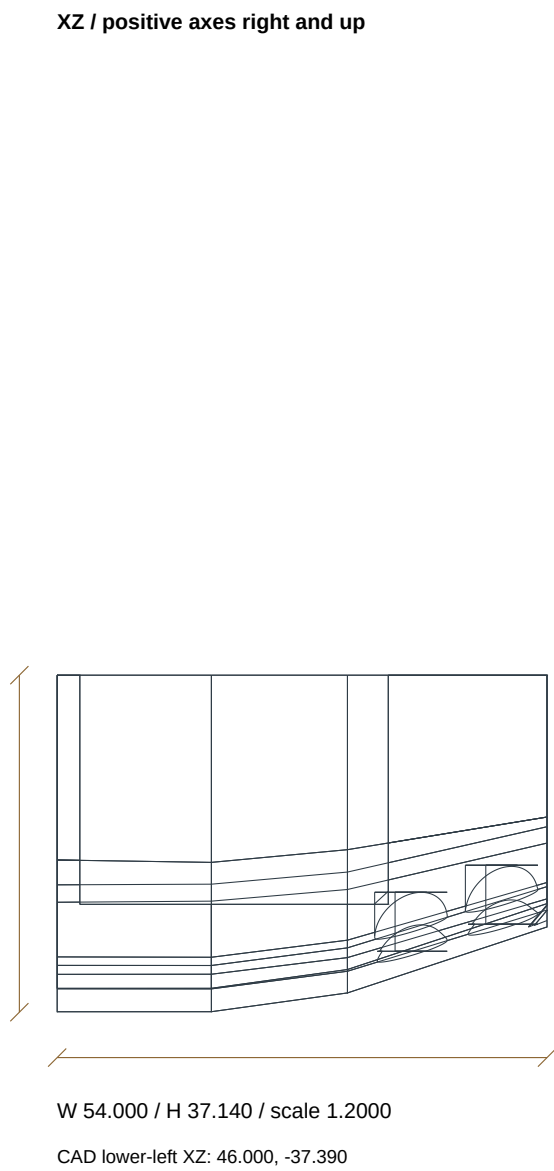
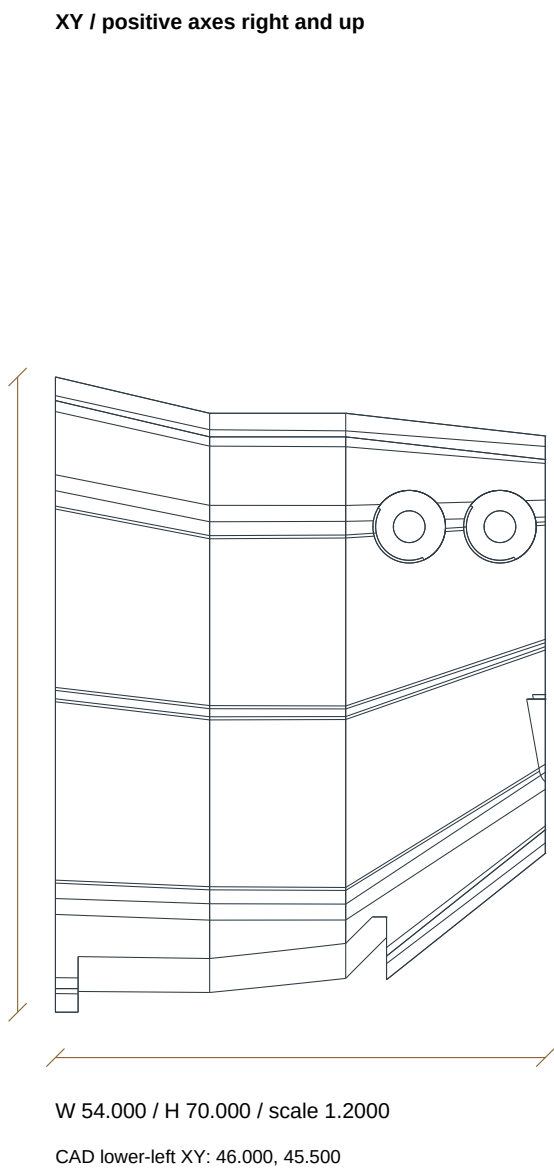
Same selected long swept outline; original fixing seats retained.  
End plane X119 in J4.rotor; nominal1mm planar seam to J5 cowl envelope startingX120.  
Local RS03 bracket clearance is based on axis offset cuts0.5mm; not a tolerance stack guarantee.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-C19-fore-ventral

Frame J4.rotor | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.



Same selected long swept outline; original fixing seats retained.  
End plane X119 in J4.rotor. Local inner relief from actual downstream carrier at J5=115..130deg.  
Shape and mounts must match current STEP; do not print an untrimmed A17 fore cover.

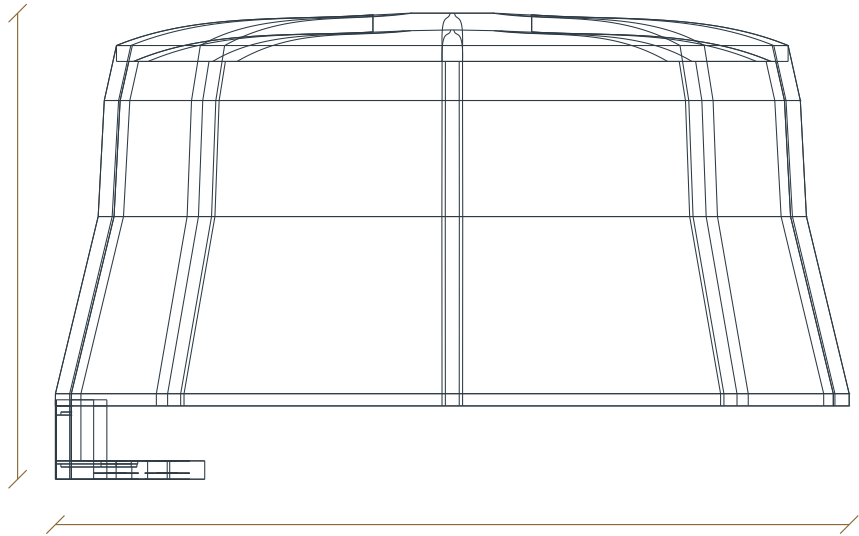
No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.  
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-C21-J5-cowl-a

Frame J5.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

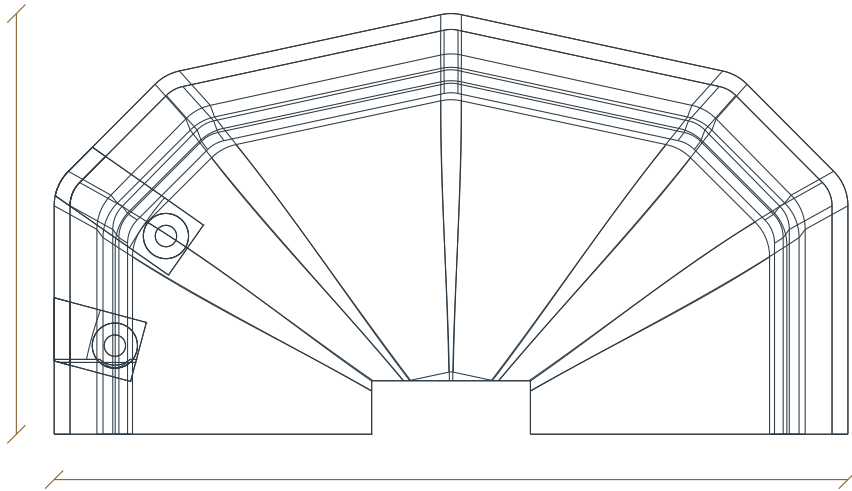
STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

XY / positive axes right and up



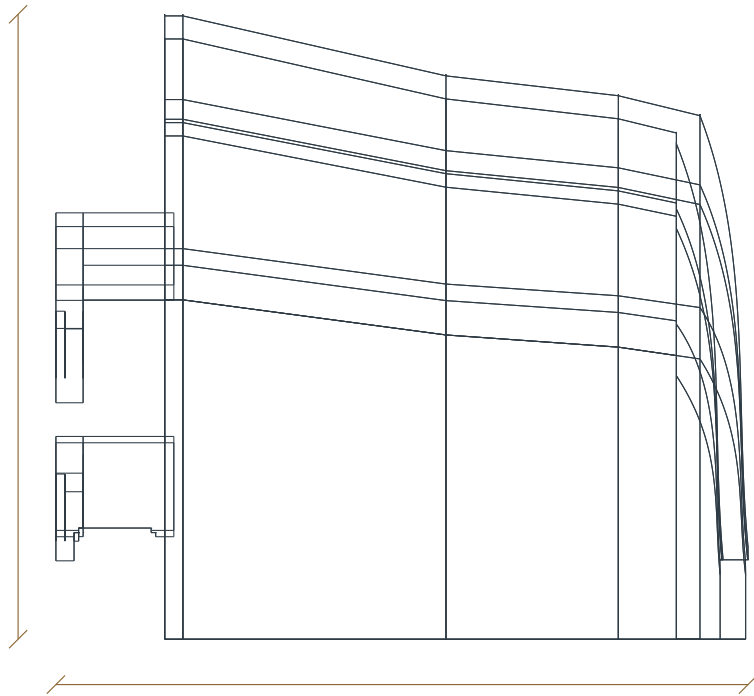
W 130.000 / H 76.300 / scale 0.8077  
CAD lower-left XY: -65.000, 31.000

XZ / positive axes right and up



W 130.000 / H 68.867 / scale 0.8077  
CAD lower-left XZ: -65.000, 0.250

YZ / positive axes right and up



W 76.300 / H 68.867 / scale 1.2000  
CAD lower-left YZ: 31.000, 0.250

Upper split curved carapace over actual RS03. Rear service window26x18 is a placeholder clearance.  
Four cover stations shared with lower cowl: R57,145/165/185/195deg; split ownership as STEP.  
DIN7984 M3x16 / nutM3 / DIN125 washers; counterboreD7.4 depth1, tab3 thick, floor2.  
Low headH2 plus0.5 washer protrudes1.5 above tab; no thick flush crown in J6 swept space.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

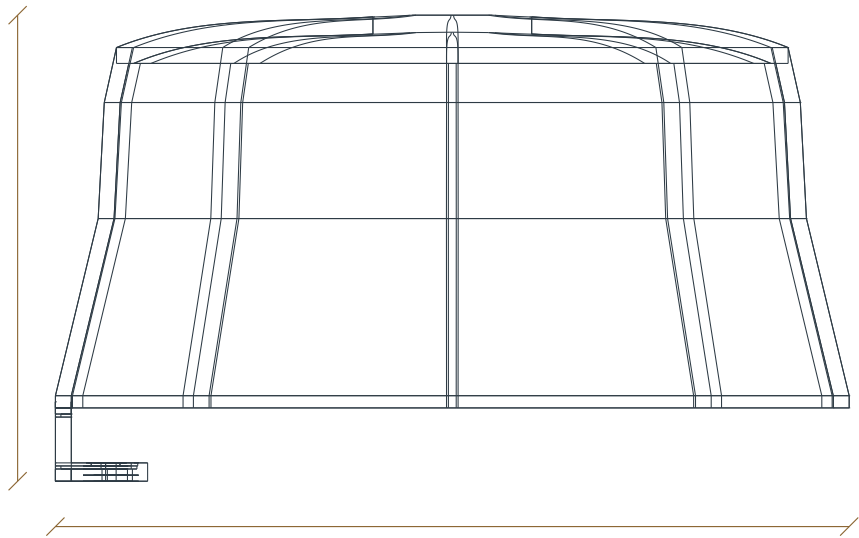
Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

A19-C21-J5-cowl-b

Frame J5.fixed | printed\_cover | mm | NOMINAL SUPPORTED FIT ISSUE

STEP defines geometry. All BREP edges shown, including hidden edges. Each view has its own stated scale.

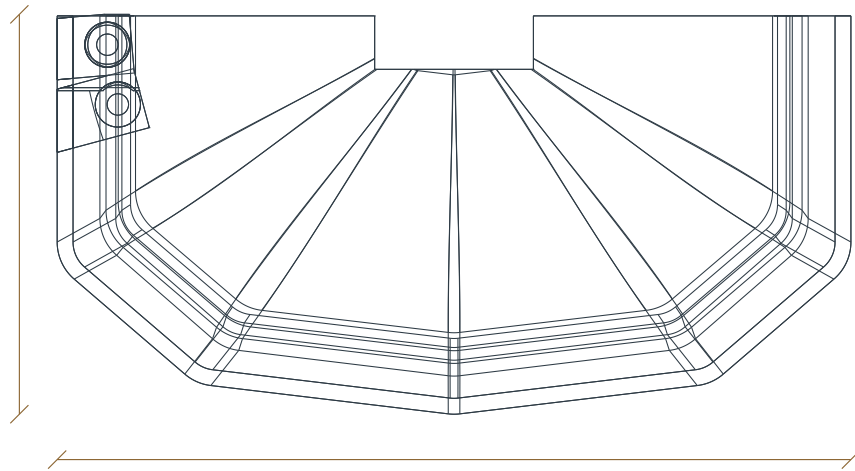
XY / positive axes right and up



W 130.000 / H 76.300 / scale 0.8077

CAD lower-left XY: -65.000, 31.000

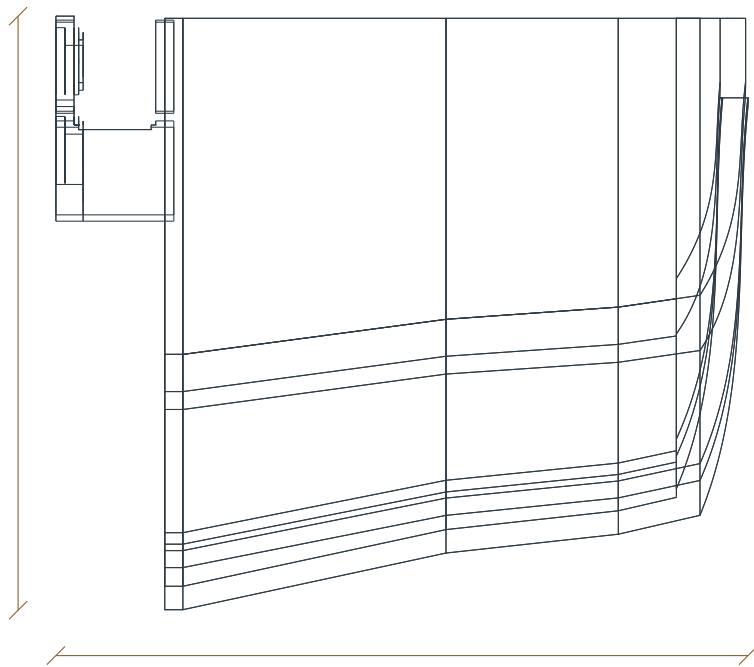
XZ / positive axes right and up



W 130.000 / H 65.463 / scale 0.8077

CAD lower-left XZ: -65.000, -65.463

YZ / positive axes right and up



W 76.300 / H 65.463 / scale 1.2000

CAD lower-left YZ: 31.000, -65.463

Lower split curved carapace over actual RS03; preserve seam and retained rear service opening.  
DIN7984 M3x16 / nutM3 / DIN125 washers; counterboreD7.4 depth1, tab3 thick, floor2.  
Nominal grip10.5 + head washer0.5; rear thread5, nut2.4, exposed2.6.  
Install rear nuts before cowls. Tool31 only checks nominal straight shaft, not real wrench/nut retention.

No metal GD&T or general tolerance released. Calibrate printer and measure real hardware before drilling critical features.

Printed structural parts require external support, all power off and no payload. Purchased steel/aluminum parts are not printed.

STEP SHA256 ab9417829c09fe804bfdbb9e... | Source assembly 4e76c049091d411b...